

# A Network Theory of Project Management\*

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August 5, 2026

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## Abstract

A principal assigns workers to projects by capping the effort they may exert. Workers differ in their preferences for projects, enjoy collaborating, but find spreading effort across projects costly. Optimal project assignments constitute a *nested hierarchy*. Each worker's assignment is derived from the same ranking of projects—the principal's marginal value in the optimum—but with two idiosyncratic cutoffs: he is unconstrained at projects above the first, capped between the two, and unassigned below the second. The principal tends to cap a worker where he most prefers to work. Shocks propagate through the network induced by the project assignment. An improvement local to one project can degrade every other project in the organization.

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\*I am indebted to my advisor Evan Sadler for his continued encouragement and advice. I thank Arslan Ali, Yeon-Koo Che, Mark Dean, Ben Golub, Melanie Chen, Navin Kartik, Tianhao Liu, Alessandro Pavan and audiences at the Columbia Microeconomic Theory Colloquium and the Economics of Networks Workshop at Stony Brook for their helpful comments.

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# 1 Introduction

Organizations routinely limit members from doing more of something both parties value. A physician holding a career development award from the National Institutes of Health cannot spend more than a quarter of his time with patients.<sup>1</sup> Engineers in product development have a cap on the number of projects they can join at one time.<sup>2</sup> An associate at a consulting firm is held below full billable utilization on a flagship project he is passionate about and staffed to work on tasks for smaller clients. In all of the above, the organization values the restricted activity, the member is willing to contribute more effort, yet the limit is administered rather than priced outright.<sup>3</sup>

Economic theory has largely focused on how organizations shape behavior through incentives. However, compensation is not the margin most organizations adjust day to day. For many public and nonprofit organizations, performance pay is unavailable due to legal or budgetary hurdles.<sup>4</sup> Even in corporations with greater contracting freedom, managers assemble project teams with little to no authority over wages. What they do instead is allocate workers to different projects, which has measurable consequences for firm performance (Minni, 2026).<sup>5</sup> Wages are not a substitute because pulling a worker back from an activity he enjoys requires a contract that *penalizes* extra effort.<sup>6</sup> This paper takes assignment seriously as a managerial tool and studies how managers—through formal limits on worker activities—organize production by deciding who works on what, how much he

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<sup>1</sup>NIH Career Development K awards stipulate a split of 75% full-time professional effort and 25% clinical work.

<sup>2</sup>As multi-project work increases, [Wheelwright and Clark \(1992\)](#) argues that worker productivity may drop because they are not able to allocate their time optimally.

<sup>3</sup>Management tools like Jira, Time Doctor and Clockify allow managers to monitor and enforce these limits.

<sup>4</sup>Under [Title 5 of the United States Code](#), the yearly monetary incentives a government employee receives cannot exceed a specific threshold. Labor unions frequently file lawsuits against the federal government's implementation of pay-for-performance. See the [American Federation of Government Employees' June 2025 memorandum](#) opposing the Office of Personnel Management's overhaul of performance reviews.

<sup>5</sup>See [Benson and Shaw \(2025\)](#) for a broader discussion of the literature on how managers create value through job allocation.

<sup>6</sup>In general, caps and wages move effort in opposite directions, and the only outcome both can implement is the efforts workers would choose on their own. I leave further discussion to [Section 6.3](#).

works, and with whom.

I develop a model where a principal (she) assigns workers (he) to projects by placing a cap on the effort each worker can exert at each project. I combine three well-documented economic forces that have been generally analyzed in isolation. First, each worker has an intrinsic preference for each project, which captures fit: his skill set, his taste, the difficulty of the project, etc. (Kristof-Brown et al., 2005; Ritz et al., 2016; Carpenter and Gong, 2016). Second, he enjoys working alongside others, and effort within a project is a strategic complement (Hoegl and Gemuenden, 2001; Mas and Moretti, 2009; Chiocchio et al., 2011). Third, effort comes at a quadratic cost governed by two parameters: a specialization cost that penalizes concentrating effort on a single project, and a multitasking cost that taxes the spread of effort across multiple projects (Zika-Viktorsson et al., 2006; Leroy, 2009; Coviello et al., 2014).

The principal values the quality of projects through an increasing and concave function and the quality of a project is given by the sum of its efforts. Formally, she chooses caps on workers' efforts and they play a Nash equilibrium in the induced effort game. I provide conditions for a unique equilibrium and a convex reformulation of the principal's problem in which she directly chooses efforts that satisfy workers' incentive constraints (Proposition 1 and 2).

Effort caps provide the principal with two distinct tools: assignment and utilization. A positive (zero) cap *(un)assigns* a worker to a project. Setting a positive cap strictly below what a worker would freely contribute *underutilizes* him. He would like to work more, but is prohibited from doing so, leaving his incentive constraint slack. Setting a cap above what he would like to contribute *fully utilizes* him. Both tools have different impacts on the network of interactions. Unassigning a worker from a project removes him as a source of collaboration. Underutilization keeps him as a source of collaboration, but the effort cap, not his incentive constraint, determines his effort at that project.

The main result is that optimal project assignments organize project teams into a *nested hierarchy*. Rank projects by the principal's marginal value at the optimum. Each worker's set of assigned and fully utilized projects is sufficiently characterized by two cutoffs  $\sigma_i \leq \gamma_i$  on that ranking. He is fully utilized at every project with a larger marginal value than  $\gamma_i$ , assigned but underutilized at every project

between  $\gamma_i$  and  $\sigma_i$ , and unassigned at projects with marginal value below  $\sigma_i$  ([Theorem 1](#)).

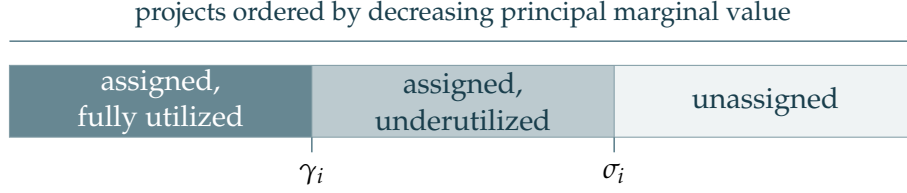


Figure 1: Optimal worker project schedule. Worker  $i$ 's two cutoffs sit on the ranking of projects by the principal's marginal value in the optimum, a ranking every worker shares.

That optimal project assignments take a simple structure is surprising given the richness in the primitives. Each of the three forces that shape a worker's effort provision— preferences, collaboration and costs— produces very different organizational structures when considered in isolation. A natural conjecture is that the principal exploits worker comparative advantages and concentrates workers at projects they prefer, forming distinct teams with strong complementarity. Assignments across projects might overlap with little to no regularity. [Theorem 1](#) states otherwise. There are no gaps in a worker's schedule: no worker is ever assigned or fully utilized at a low-value project while unassigned or underutilized at one of higher value.

The principal exercises her assignment/utilization instrument most at projects she values least on the margin. These need not correspond to projects she weights least in her valuation function. A project with high weighting could draw ample effort from many workers who enjoy working on it, causing its marginal value to fall below that of another project with a nominally lower weight.<sup>7</sup> She caps workers at the former to redistribute effort towards projects with higher gains on the margin. The example of the consulting associate in the opening paragraph is a leading case. If a worker is assigned to two projects  $p$  and  $q$  and prefers  $p$  substantially more than  $q$ , then the principal must be underutilizing him at  $p$  ([Proposition 4](#)). Paradoxically, the principal caps a worker precisely where he prefers to contribute most.

While [Theorem 1](#) provides structure, it does not predict where these cutoffs lie because the ranking of projects is endogenous. To give sharper insights, I assume

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<sup>7</sup>[Example 1](#) is such a case.

the principal’s objective is Cobb-Douglas and vary worker attributes one dimension at a time while holding the others fixed. In each of the following environments, the marginal value ordering of projects in the optimum aligns with their exogenous Cobb-Douglas weights.

When workers are identical, the assignment margin disappears because every worker works on every project in a worker-symmetric optimal project assignment. A single project index divides the ranking: every worker is fully utilized at projects above it and underutilized at projects below it ([Theorem 2](#)). With higher multitasking costs, stronger complementarities, or a larger worker pool, the number of projects with fully utilization shrinks. Concentrating effort towards projects at the top is more beneficial, and the principal caps workers on more projects at the bottom to force this imbalance. Higher specialization costs expand the number of fully utilized projects for the opposite reason.

Reintroducing heterogeneity turns the assignment margin back on, and the question of interest is which attributes sort workers. Multitasking costs alone sufficiently sort workers ([Theorem 3](#)). Workers with high multitasking costs specialize on only the most valuable projects, while workers with low multitasking costs serve as generalists and are assigned a wide range of projects. The extensive margin is ordered while the intensive margin is not. Whether a specialist works more or less than a generalist on a project they share is sensitive to the parameters.

When workers differ in how well they collaborate, it is natural to believe that highly central workers, because they are valuable to many project teams, should be spread across them. However, standard network game statistics like Katz-Bonacich centrality get the ordering wrong. Designating a central worker to be a generalist *dampens* his contributions everywhere due to multitasking costs. The same collaboration links that make a worker valuable precisely make it costly to spread his efforts across them. The correct sorting statistic, which I call a *congestion index*, that captures this trade-off turns out to be a ratio of two Katz-Bonacich centralities with different weightings ([Theorem 4](#)). Moreover, the relevant network of interaction shrinks as we descend the project ordering. Once a specialist is unassigned from a project, he is no longer a source of complementarity for any worker left at that project, and so congestion indices are computed recursively on a series of shrinking networks.

Fixing an optimal project assignment, I derive its comparative statics. Through effort caps, the principal chooses which links are active and the magnitude of complementarity generated in equilibrium. In essence, the statics are those of an endogenous network, and they differ depending on the pattern of assignment and utilization.

When every worker is assigned to and fully utilized at every project, any shock decomposes into a within-project spillover and a network reallocation effect ([Theorem 5](#)). For intuition, consider raising worker  $i$ 's preference for working on project  $p$ . Collaboration encourages every worker to work more on project  $p$ , but higher efforts raise total efforts, and subsequently, marginal costs. Workers withdraw from other projects, which is compounded by a loss in complementarity there. A positive shock local to one project degrades the quality of all other projects within the organization.

When the principal keeps some workers off some projects, the comparative statics become more nuanced. How the network reallocation effect manifests depends on the particular route a shock travels through ([Theorem 6](#)). An imbalance in complementarities stemming from imbalanced assignments may lead a worker to reduce more effort at one project than he adds at another, dragging his marginal costs down. This frees up capacity for him to raise efforts at another project not subject to the original shock. In contrast to the previous result, a positive shock could *increase* the quality of another project.

When the optimal project assignment features underutilization, the comparative statics behave differently yet again. Previously, efforts and shock propagation were entirely pinned down by the incentive constraints because of full utilization. With underutilization, the principal has direct control over efforts at slack incentive constraints, enabling her to substitute one worker's effort for another whenever a shock obliges her to do so. Shocks admit a Slutsky decomposition into an income and substitution effect ([Theorem 7](#)). The income effect decomposes into the aforementioned within-project spillover and network reallocation effects, while the magnitude of the substitution effect is determined by the curvature of the principal's valuation.

## 1.1 Related Literature

This paper contributes to three areas of research: network games, principal-agent theory and organization design.

**Network games.** Early theories of network games restrict workers to choosing a single action with either complementary or substitutable interactions between neighbors (Ballester et al., 2006; Bramoullé and Kranton, 2007; Bramoullé et al., 2014). Two recent papers, Claveria-Mayol et al. (2024) and Dasaratha et al. (2025) embed a network game within a moral-hazard problem and characterize optimal contracts. My paper differs along three dimensions. First, the principal in both papers above faces a single project; in my model, she divides worker efforts across multiple projects. Second, rather than incentive provision, the principal in my model uses effort caps as her instrument of choice. Third, both papers implicitly take the set of active agents as given. My analysis explicitly characterizes the set of active agents i.e. the assigned agents in a multi-project setting.

Work on network interventions posits a social planner who removes players or designs a vector of marginal benefits to influence activity (Belhaj and Deroian (2018); Galeotti et al. (2020, 2025)). The principal in my model instead controls workers' action spaces. By determining how much complementarity workers transmit within a project, she effectively selects which links within a network are active, and to what extent. The network is designed, whereas in the network formation literature, link formation is decentralized (e.g. Jackson and Wolinsky (1996); Bala and Goyal (2000)).

This paper directly builds on network games in which players choose multiple actions. In multiplex network games, players act against each other across several layers subject to an aggregate effort constraint (Chen et al. (2018); Zenou and Zhou (2026)).<sup>8</sup> Closest to this work is Sadler and Yeh (2026), in which players select neighbor-specific efforts subject to a convex cost of total effort. Workers in my model choose project-specific efforts which exhibit strategic complementarities and positive spillovers within a project. The effort game is a parametric multi-graph version of Sadler and Yeh (2026). The novelty is that the principal selects the

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<sup>8</sup>Applications studied include conflict (Franke and Öztürk, 2015; Xu et al., 2022) and communication networks (Demange, 2025)).



structure of the effort game, and the comparative statics pertain to an endogenous network. In a separate project, I study the optimality of modular task assignments where effort at a task provides positive spillovers but are strategic *substitutes* for locally related tasks, which introduces undesirable free-riding in teams (Yeh, 2026).

**Principal-agent theory.** Models of principal-agent problems have focused on incentive design in multitask or multi-agent settings (Itoh, 1991; Holmstrom and Milgrom, 1991; Dewatripont et al., 2000; Che and Yoo, 2001). This paper combines both dimensions and replaces the contracting instrument with assignments.

Within this literature, assignment is most closely related to delegation. Setting an effort cap is akin to choosing a delegation set (Alonso and Matouschek, 2008), as applied in project choice (Armstrong and Vickers, 2010) and multidimensional actions (Frankel, 2014). I highlight three differences. First, delegation is traditionally motivated by asymmetric information. The principal balances granting an agent flexibility to use his private information while limiting undesirable actions resulting from agent bias. My setting has no private information. Restricting action sets is valuable to the principal because effort is scarce due to convex costs; forbidding effort at one project allows the principal to obtain more effort at another. Second, the nature of actions that are excluded is reversed. In delegation, the principal oftentimes restricts the agent from taking actions the principal dislikes; in my model, the principal precludes a worker from taking actions both enjoy. Third, the delegation sets in my environment are interdependent. Adjusting a worker's effort cap at one project necessarily affects his action set at another project through substitutability and other workers' action sets at the same project through complementarity.

**Organization design.** With transferable utility and output that is supermodular in agents' types, the classic result by Becker (1973) finds the optimal assignment to be positively assortative. In most settings, agents are sorted by a single scalar attribute (e.g. Kremer (1993)). The richness of my environment prevents the principal from sorting agents by any single attribute. A worker who is very good at multitasking could be a poor teammate, and it is not immediately clear which workers strongly complement each other on different projects. Related to this paper is Freund (2025), who show that specialization within a firm generates coworker complementarity and induces a concentration of talent within a small number of firms. In their



model, workers belong to a single firm, and so the question of multi-team membership is moot.<sup>9</sup> In my model, a worker may be assigned to several projects at once, and multitasking costs and complementarity drive the nested structure of project teams.

The resulting structure connects to optimal hierarchies, which arise from limited managerial attention (Geanakoplos and Milgrom, 1991), limited decision-making capacity (Hart and Moore, 2005), communication (Visser, 2000; Ambrus et al., 2013) and grouping activities for synergy (Dessein et al., 2010). These are hierarchies of authority or information. In contrast, the nested hierarchy in an optimal project assignment is a hierarchy of membership ordered by project marginal values.

## 2 Model

There is one principal (she), a set of workers (he)  $N = \{1, \dots, n\}$  and a set of projects  $P = \{1, \dots, m\}$ .

**Workers.** Worker  $i$  chooses nonnegative effort  $e_i^p \in [0, \bar{e}_i^p]$  to play at each project  $p$  where  $\bar{e}_i^p > 0$ . He receives utility given by the sum of per-project benefits less his total cost of effort.

$$U_i(\mathbf{e}_i, \mathbf{e}_{-i}) = \underbrace{\sum_{p=1}^m (\alpha_i^p e_i^p + \sum_{j \neq i} g_{ij} e_i^p e_j^p)}_{\text{per-project benefit}} - \underbrace{\left( \sum_{p=1}^m \frac{c_i}{2} (e_i^p)^2 + r_i \sum_{1 \leq p < q \leq m} e_i^p e_i^q \right)}_{\text{cost of effort}}$$

where  $\mathbf{e}_i = \{e_i^p\}_{p=1}^m$  and  $\mathbf{e}_{-i}$  represents all other workers' efforts at all projects.

At each project  $p$ , worker  $i$  receives a benefit that is linear in his effort.  $\alpha_i^p > 0$  is his intrinsic preference for working on project  $p$ .  $g_{ij} > 0$  measures the level of effort complementarity between worker  $i$  and  $j$ , which I assume to be symmetric i.e.  $g_{ij} = g_{ji}$ .<sup>10</sup>

Worker  $i$  incurs a general quadratic cost of effort.  $c_i > 0$  is his cost of specialization

<sup>9</sup>For additional work on the optimal composition of a team and how it is affected by size, diversity, and incentives, see Prat (2002); Georgiadis (2015); Fu et al. (2016); Glover and Kim (2021).

<sup>10</sup>The complementarity terms  $g_{ij} e_i^p e_j^p$  can be equivalently interpreted as "helping" efforts by colleagues that reduce a worker's cost of effort instead of raising his benefit from working.

on a single project.  $r_i > 0$  measures his cost of multitasking, which penalizes the spread of his efforts across multiple projects. To maintain convexity in costs, I assume that  $c_i > r_i$ .<sup>11</sup> Because costs are convex and benefits are linear in a worker's own efforts, workers' utilities are concave in their own efforts.

**Principal.** The quality of project  $p$  is given by the sum of worker efforts  $\mathcal{E}^p := \sum_{i=1}^n e_i^p$ . The principal values project qualities according to  $V : \mathbb{R}^m \rightarrow \mathbb{R}$ , which is strictly increasing in each argument, strictly concave and  $C^2$ . The principal's choice of upper bounds  $\bar{\mathbf{e}} \in \mathbb{R}_+^{mn}$  solves:

$$\max_{\bar{\mathbf{e}} \geq 0} V(\mathcal{E}^1, \dots, \mathcal{E}^m)$$

s.t.  $\mathbf{e}$  is the (principal-preferred) pure-strategy Nash equilibrium given  $\bar{\mathbf{e}}$

If there are multiple equilibria induced by  $\bar{\mathbf{e}}$ , then the one that achieves the highest principal valuation is selected.

## 2.1 Implementability

In this section, I characterize the feasible set of the principal's problem. Symmetric effort complementarity permits us to write the workers' effort game as an exact potential game. The exact potential is

$$\Phi(\mathbf{e}) = \sum_{i=1}^n U_i(\mathbf{e}) - \frac{1}{2} \sum_{i=1}^n \sum_{j \neq i} \sum_{p=1}^m g_{ij} e_i^p e_j^p.$$

**Assumption 1.** *The interaction matrix, given by*

$$\mathbf{H} := (\mathbf{C} \otimes \mathbf{I}_m + \mathbf{R} \otimes (\mathbf{1}_m \mathbf{1}_m^\top - \mathbf{I}_m)) - (\mathbf{G} \otimes \mathbf{I}_m)$$

*is positive definite, where  $\mathbf{C} := \text{diag}(c_i)_{i \in N}$ ,  $\mathbf{R} := \text{diag}(r_i)_{i \in N}$ ,  $\mathbf{G}$  has entries  $\mathbf{G}_{ij} = g_{ij}$  and 0 on its diagonal, and  $\otimes$  is the Kronecker product.*<sup>12</sup>

<sup>11</sup>The Hessian matrix of worker  $i$ 's cost of effort is  $(c_i - r_i)\mathbf{I}_m + r_i \mathbf{1}_m \mathbf{1}_m^\top$ , which has eigenvalues  $c_i + (m-1)r_i$  and  $c_i - r_i$  with multiplicity of  $m-1$ . For the cost to be strictly convex, its Hessian needs to be positive definite, which is true if  $c_i - r_i > 0$ .

<sup>12</sup>Formally, given a  $m \times n$  matrix  $\mathbf{A}$  and a  $p \times q$  matrix  $\mathbf{B}$ , the Kronecker product  $\mathbf{A} \otimes \mathbf{B}$  is a

Under this assumption, the Hessian of the exact potential is  $-\mathbf{H}$ , and so the exact potential is strictly concave. It admits a unique maximizer corresponding to the unique Nash equilibrium of the workers' effort game.<sup>13</sup>

$\mathbf{H}$  is a large  $nm \times nm$  matrix with worker-project pairs for indices. Each element captures an interaction between two worker-project pairs. Working on project  $q$  raises worker  $i$ 's marginal cost at project  $p$  by his multitasking parameter  $r_i$ , recorded in  $\mathbf{H}[(i, p), (i, q)] = r_i$ . The same goes for specialization costs. When worker  $j$  increases his effort on project  $p$ , it raises worker  $i$ 's marginal benefit at project  $p$ <sup>14</sup> through collaboration, represented by  $\mathbf{H}[(i, p), (j, p)] = -g_{ij}$ . There are no explicit interactions between worker  $i$  at project  $p$  and worker  $j$  at project  $q$  i.e.  $\mathbf{H}[(i, p), (j, q)] = 0$  because synergies between workers operate within a project, and multitasking costs affect a single worker across multiple projects. Interactions between different workers at different projects surface in equilibrium.

Uniqueness of the Nash equilibrium guarantees that no selection rule is required in the principal's problem. The following result ensures it is without loss for the principal to directly choose efforts so long as each worker's *incentive constraints* are respected.

**Proposition 1.** *For  $\mathbf{e} \geq \mathbf{0}$ , there exists an upper bound  $\bar{\mathbf{e}}$  that implements  $\mathbf{e}$  as an equilibrium if and only if*

$$\alpha_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q$$

*whenever  $e_i^p > 0$ . Moreover, the principal can implement  $\mathbf{e}$  by setting  $\bar{\mathbf{e}} = \mathbf{e}$ .*

*Proof.* See Appendix A.1. □

When a worker's marginal utility from working on a project is nonnegative,

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$$mp \times nq \text{ block matrix given by } \mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ \vdots & \ddots & \vdots \\ a_{m1}\mathbf{B} & \cdots & a_{mn}\mathbf{B} \end{bmatrix}.$$

<sup>13</sup>There are other ways of showing a Nash equilibrium exists and is unique. One is to write down each worker's best response function and then appeal to Brouwer's Fixed Point Theorem. Uniqueness follows from applying the concave games result in Rosen (1965). The fixed-point iteration does not provide a practical approach to computing the Nash equilibrium. Another is to cast the equilibrium as the optimum of a constrained quadratic program, where the quadratic form is the exact potential.

<sup>14</sup>Equivalently, it lowers worker  $i$ 's marginal cost for project  $p$ .

reducing effort is never profitable. Setting the effort cap to be exactly the desired effort level prevents the worker from contributing more even when his marginal utility is strictly positive. I call the condition outlined in [Proposition 1](#) worker  $i$ 's **incentive constraint** at project  $p$ . Due to the linear-quadratic framework, each incentive constraint is linear in worker efforts and defines a closed half-space.

The left term is worker  $i$ 's *effective* marginal benefit at project  $p$ . It depends on his intrinsic preference and the efforts of other workers at that project. An increase in other workers' efforts at project  $p$  relaxes his incentive constraint. The principal can encourage or discourage effort from worker  $i$  at project  $p$  by raising or lowering his effort cap. The right term is worker  $i$ 's *effective* marginal cost at project  $p$ . To encourage him to work more on project  $p$ , raising his effort cap is not enough if his incentive constraint binds. The principal has to reduce his efforts at other projects by lowering effort caps, or by encouraging more effort from other workers at that project.

## 2.2 Managerial Tools: Assignment and Utilization

By controlling effort caps, the principal not only directs efforts towards projects she cares about, but also shapes the pattern of interaction among workers.

**Definition 1.** Given effort profile  $\mathbf{e}$ , worker  $i$  is **assigned** to project  $p$  if  $e_i^p > 0$ , otherwise he is **unassigned**. Worker  $i$  is **fully utilized** at project  $p$  if his incentive constraint binds i.e.

$$\alpha_i^p + \sum_{j \neq i} g_{ij} e_j^p = c_i e_i^p + r_i \sum_{q \neq p} e_i^q.$$

He is **underutilized** if

$$\alpha_i^p + \sum_{j \neq i} g_{ij} e_j^p > c_i e_i^p + r_i \sum_{q \neq p} e_i^q.$$

To see how assignment and utilization fundamentally alter how workers interact, consider the environment where workers are free to choose efforts as they wish. Every worker works at every project up until their incentive constraints bind, and their best responses recorded in the interaction matrix  $\mathbf{H}$  pin down the

unique equilibrium (assuming it is interior),

$$\hat{\mathbf{e}} := (\mathbf{H})^{-1}\boldsymbol{\alpha}.$$

Unassigning worker  $i$  from project  $p$  nullifies  $(i, p)$  as a source of any interactions. That is, worker  $i$  at project  $p$  generates no complementarity for others at  $p$  or substitutability for his effort at other projects. The principal removes both the  $(i, p)$ -th row *and* column of  $\mathbf{H}$ , revealing a square principal submatrix of  $\mathbf{H}$ . The new equilibrium can differ significantly as submatrix inversion cannot be predicted from the full matrix inversion.

Through underutilization, the principal deliberately keeps the  $(i, p)$  incentive constraint slack. She removes the  $(i, p)$ -th row of  $\mathbf{H}$ , but keeps the  $(i, p)$ -th column. The  $(i, p)$ -th row tracks how efforts throughout the organization contribute to worker  $i$ 's incentive constraint at project  $p$ . Once the incentive constraint is slack, it is no longer relevant for determining the equilibrium effort worker  $i$  plays at project  $p$ . By contrast, the  $(i, p)$ -th column remains intact because worker  $i$ 's effort at project  $p$  still affects other incentive constraints through collaboration and multitasking spillovers. Underutilization creates a rectangular matrix. How an underutilized worker interacts with others, which was originally determined by the row now removed, is instead governed by the principal's choice of effort cap.

Utilization also informs the principal how much latitude she has when setting effort caps. Different effort caps may implement the same effort profile as their unique equilibrium. If worker  $i$  is fully utilized at project  $p$  for  $\mathbf{e}$ , then any effort cap  $\bar{e}_i^p \geq e_i^p$  is sufficient. In effect, the principal can leave him *unconstrained* at project  $p$ . However, for a worker  $i$  who is underutilized at project  $p$ , the principal must set a binding cap  $\bar{e}_i^p = e_i^p$ . Setting a higher effort cap induces worker  $i$  to increase his equilibrium effort at project  $p$ . A different equilibrium realizes because  $i$  necessarily shirks at other projects in order to work more on  $p$ .

### 2.3 A Convex Optimization Problem

There is a slight technical complication concerning assignment. By setting a zero effort cap, a worker must play zero effort even if his marginal utility is neg-

ative— incentive constraints are effectively ignored at zero effort. Keeping track of which incentive constraints are active, which depends on whether effort is positive or zero, requires a separate binary variable. The feasible set is nonconvex, and the combinatorial difficulty associated with a binary optimizer makes the problem intractable.<sup>15</sup> I focus on a family of model parameters under which incentive constraints are never violated at zero effort, which implies that incentive constraints *always* need to be satisfied.

**Proposition 2.** Define  $\mathbf{P} := \mathbf{C} + (m - 1)\mathbf{R} - \mathbf{G}$  and  $\tilde{\mathbf{E}} := \mathbf{P}^{-1} \sum_{p=1}^m \alpha^p$ . If  $r_i \tilde{\mathbf{E}}_i \leq \min_p \alpha_i^p$  for all  $i$ , then the principal's problem is given by the following convex optimization problem:

$$\begin{aligned} & \max_{\mathbf{e}} V(\mathcal{E}^1, \dots, \mathcal{E}^m) \\ & \text{s.t. } \mathbf{e} \geq \mathbf{0}, \\ & \forall (i, p), \alpha_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q \\ & \forall p, \sum_{i=1}^n e_i^p = \mathcal{E}^p \end{aligned}$$

When workers are homogeneous, the condition reduces to  $r + g(n - 1) \leq c$ , which is implied by [Assumption 1](#).

*Proof.* See [Section A.2](#). □

The incentive constraint for worker  $i$  at project  $p$  with zero effort ( $e_i^p = 0$ ) is

$$r_i \sum_{q \neq p} e_i^q \leq \alpha_i^p + \sum_{j \neq i} g_{ij} e_j^p$$

The proof shows that  $\tilde{\mathbf{E}}_i$  bounds the total effort worker  $i$  can exert at any implementable effort profile. The condition imposes no additional restrictions when workers are homogeneous. If multitasking costs are not too high relative to specialization costs, then the incentive constraint always holds at zero effort. With this, we can treat positive and zero efforts the same and incentive constraints are always

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<sup>15</sup>For more details on formulating the general principal's problem as a mixed-integer nonlinear program, see [Appendix C](#).

required in the optimum. The principal's problem is a tractable convex optimization problem which I focus on for the rest of the paper. A solution exists because the objective is continuous and the feasible set is closed and bounded. Moreover, the solution is unique in *total* efforts because the objective is strictly concave over total efforts and the feasible set is convex.

## 2.4 Remarks

1. Symmetry in complementarity is sufficient for a unique equilibrium. Asymmetric complementarities, discussed in [Section 6.1](#), do not affect the main structural results.
2. There are no explicit contracts in this model. Worker  $i$ 's intrinsic preference for project  $p$ ,  $\alpha_i^p$ , reflects his personal motivation, taste, skill set, talent, project difficulty, etc. In [Section 6.3](#), I consider an alternate model where the principal chooses transfers. To achieve underutilization without an effort cap, the principal needs to implement *negative* transfers, which is barred by limited liability.
3. Imposing an external budget on effort caps does not change the main structural results as I show in [Section 6.2](#).
4. The principal is not a social planner and her objective does not account for the effort costs borne by the workers. Rather, they shape the feasible set in her optimization problem.  $V$  can be interpreted as a firm production function that exhibits diminishing returns due to concavity. I discuss welfare implications of effort caps in [Section 6.4](#).
5. The participation constraint  $U_i \geq 0$  is trivially satisfied given that implementability requires that marginal utility is nonnegative and the fact that marginal utility is diminishing in own efforts. If workers face a positive outside option  $U_i \geq \underline{U}_i > 0$ , then the analysis becomes intractable as the feasible set is nonconvex.



### 3 Structure of Optimal Project Assignments

#### 3.1 Geometric Alignment

Before detailing the main structural result, I first provide a geometric characterization of optimal project assignments. For notation, I write  $V^p := \frac{\partial V}{\partial \varepsilon^p}$ .

**Proposition 3.** *Let  $\mathbf{e}$  be optimal and define  $A^p := \{i \mid e_i^p > 0\}$ . There exists  $\boldsymbol{\lambda} \geq \mathbf{0}$  such that*

$$V^p = (\mathbf{H}\boldsymbol{\lambda})_i^p \text{ if } (i, p) \in A^p \quad \text{and} \quad V^p \leq (\mathbf{H}\boldsymbol{\lambda})_i^p \text{ if } (i, p) \notin A^p$$

*Moreover, if  $\mathbf{e} = \hat{\mathbf{e}}$ , meaning every worker is assigned and fully utilized at every project, then  $\nabla_{\mathbf{e}} V$  lies in the convex cone generated by the columns of  $\mathbf{H}$ .*

*Proof.* See Appendix A.3. □

Through a combination of (un)assignment and (under)utilization, the principal adjusts workers' efforts so that they align her valuation on the margin, represented by the gradient of  $V$ , with the columns of  $\mathbf{H}$ . In any direction she would like to push more effort, she is stopped by a binding incentive constraint. The multiplier  $\lambda_i^p$  records the shadow value of loosening worker  $i$ 's incentive constraint at project  $p$  for the principal.

This result also provides a simple diagnostic of when the principal should unassign/underutilize workers. If  $(\mathbf{H})^{-1} \nabla_{\mathbf{e}} V \geq 0$  at  $\hat{\mathbf{e}}$ , then no adjustment of effort caps is required; simply let the workers choose efforts as they desire.<sup>16</sup> On the other hand, if  $(\mathbf{H})^{-1} \nabla_{\mathbf{e}} V$  at  $\hat{\mathbf{e}}$  yields a negative entry, then the principal must unassign and/or underutilize workers to align their efforts with her valuation in the optimum.

#### 3.2 Nested Hierarchies

The main structural result requires two conditions of the optimum, which I show to be generic in Appendix B.

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<sup>16</sup>KKT conditions are both necessary and sufficient in the convex optimization problem.

**Assumption 2.** In the optimum, (1) every binding incentive and nonnegativity constraint has strictly positive multipliers and (2) worker  $i$ 's incentive constraint at project  $p$  is slack whenever  $e_i^p = 0$ .

Importantly, the converse of (2), which would eliminate underutilization from being a relevant tool, is not generically true.<sup>17</sup> The multipliers  $\lambda_i^p$  (incentive constraint) and  $\mu_i^p$  (nonnegativity) classify worker-project pairs into three distinct categories.

$$\underbrace{\lambda_i^p > 0, \mu_i^p = 0}_{\text{assigned, fully utilized}} \quad \underbrace{\lambda_i^p = 0, \mu_i^p = 0}_{\text{assigned, underutilized}} \quad \underbrace{\lambda_i^p = 0, \mu_i^p > 0}_{\text{unassigned}}$$

**Theorem 1.** Define  $A^p := \{i \mid e_i^p > 0\}$  and  $\mathcal{F}^p := \{i \mid i \text{ is fully utilized at } p\}$ . For every worker  $i$ , there exist cutoffs  $0 \leq \sigma_i \leq \gamma_i$  such that

$$i \in A^p \iff V^p \geq \sigma_i \quad i \in \mathcal{F}^p \iff V^p \geq \gamma_i$$

If projects are ordered by their marginal value to the principal in the optimum

$$V^1 \geq \dots \geq V^m,$$

then  $A^p$  and  $\mathcal{F}^p$  obey the same project ordering by set inclusion

$$A^1 \supseteq \dots \supseteq A^m \text{ and } \mathcal{F}^1 \supseteq \dots \supseteq \mathcal{F}^m.$$

*Proof.* See Appendix A.4. □

Optimal project assignments are simple. For each worker, two thresholds determine which project he is assigned to and fully utilized at. Importantly, his assignment schedule comprises three successive regimes. For the most valuable projects on the margin, the principal sets relatively high effort caps so that he is working as much as he is willing to. For medium-value projects, the worker is assigned but capped. He would like to contribute more, but the principal prohibits him

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<sup>17</sup>Underutilization is not a knife-edge phenomenon. It is defined by two *strict* inequalities, and since every incentive constraint binds only at  $\hat{e}$ , the optimum features underutilization whenever the diagnostic in the previous section yields a negative entry. [Example 1](#) has a unique optimum. Hence, the underutilization pattern there survives small perturbations of every parameter.

from doing so. On the remaining low-value projects, he is unassigned. All worker schedules agree on the *same* ranking of projects, namely by the principal's marginal value in the optimal. Project teams are necessarily nested in the optimum.

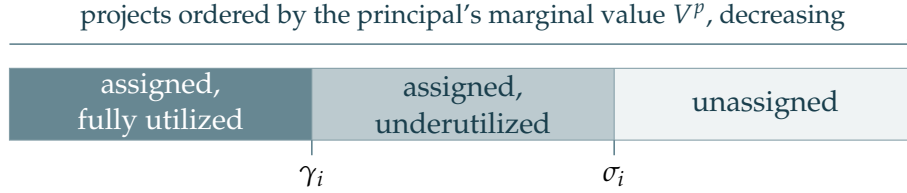


Figure 2: Optimal Worker Project Schedule

That optimal project assignments embody such a clean structure is surprising, especially because all parameters—preferences, synergies, and costs—are heterogeneous. We expect workers to retain comparative advantage at certain projects over others, and it seems natural that workers specialize in distinct projects and have non-overlapping project assignments. [Theorem 1](#) precludes any worker from being capped or excluded from a project while being fully utilized or assigned to a more valuable one. Why then should assignments nest in the optimum? The intuition lies in inspecting the stationarity condition. In optimality, the following equality obtained from rearranging the stationarity condition must hold

$$(c_i - r_i)\lambda_i^p - \mu_i^p = \underbrace{V^p + \sum_{j \neq i} g_{ij}\lambda_j^p}_{\text{shadow benefit}} - \underbrace{r_i\Lambda_i}_{\text{shadow cost}}$$

where  $\Lambda_i := \sum_{p=1}^m \lambda_i^p$  is worker  $i$ 's aggregate shadow value and captures the value to the principal of relaxing all of his incentive constraints simultaneously.

The right side is worthy of discussion. The shadow benefit term is the total value the principal gains from an extra unit of worker  $i$ 's effort at project  $p$ : the direct value of an increase in the quality of project  $p$  *and* the increased complementarity generated for his collaborators there. Worker  $i$ 's effort at  $p$  relaxes other workers' incentive constraints at  $p$  by  $g_{ij}$ , valued at  $\lambda_j^p$  by the principal. The shadow cost term takes into account the increased multitasking burden on all of worker  $i$ 's projects. His effort at  $p$  tightens his incentive constraints at another project  $q$  by  $r_i$ , valued at  $\lambda_i^q$  by the principal.

How the shadow benefit and cost for worker  $i$  at project  $p$  compare directly governs utilization and assignment. If the shadow benefit strictly exceeds the shadow cost, then he must be assigned and fully utilized at  $p$ . If they are equal, then he is necessarily assigned but underutilized at  $p$  because  $\lambda_i^p$  and  $\mu_i^p$  must both vanish.<sup>18</sup> If the shadow benefit is less than the shadow cost, then he has to be left off project  $p$ .

Two additional observations give the nested hierarchical structure. First, shadow benefits decrease as we go down the project ranking. By definition,  $V^p$  declines. That the shadow values of worker incentive constraints ( $\lambda_i^p$ ) also move in the same direction is not obvious; the intuition stems from complementarity within a project. For a more valuable project, the principal demands more effort from worker  $j$ , and this also makes worker  $i$ 's effort *more* valuable precisely because  $i$ 's effort makes  $j$ 's effort cheaper to supply. For this reason, the shadow values of incentive constraints must decrease in tandem with project marginal value. Second, the cost term carries a worker-specific term  $r_i \Lambda_i$ . It prices his effort across all projects, which is immaterial to making comparisons across projects. Putting the two together gives us the worker project schedule depicted in [Figure 2](#). Going down the project ranking reduces the shadow benefit until we reach a cross-over point where it is equal to and strictly less than the shadow cost, producing the corresponding cutoffs  $\gamma_i$  and  $\sigma_i$ .<sup>19</sup>

The simple organizational network induced by an optimal project assignment is a *nested hierarchy*. The graph will prove helpful in understanding the comparative statics of [Section 5](#).

**Definition 2.** A graph with doubly-indexed vertices  $(i, p)$  is a **nested hierarchy** if there exists an ordered partition of vertices  $\mathcal{C} = \{\mathcal{C}^1, \dots, \mathcal{C}^m\}$  such that:

1. All vertices within  $\mathcal{C}^p$  have distinct first-coordinates and form a clique.
2. Vertices sharing the same first coordinate  $(i, \cdot)$  form a clique.

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<sup>18</sup>This could occur at multiple projects as [Example 1](#) illustrates.

<sup>19</sup>The following perturbation argument is unavailable: if a worker is fully utilized or assigned to a low-value project but underutilized or unassigned to a high-value project, then shift his effort towards the high-value project. Reducing a worker's efforts at a project lowers the complementarities received by other workers, tightening and potentially violating their incentive constraints. The argument presented here sidesteps that difficulty through working with shadow values, rather than efforts directly.

3.  $(i, p) \in \mathcal{C}^p$  implies  $(i, q) \in \mathcal{C}^q$  for all  $q \leq p$ .

**Corollary 1.** Define  $G_A$  as the assignment graph with doubly-indexed vertex set  $\bigcup_p \{(i, p) \mid i \in A^p\}$ . Similarly,  $G_{\mathcal{F}}$  is the full utilization graph with doubly-indexed vertex set  $\bigcup_p \{(i, p) \mid i \in \mathcal{F}^p\}$ . In both graphs, an edge is present between two vertices whenever they share the same worker or project coordinate. In an optimal project assignment,  $G_A$  and  $G_{\mathcal{F}}$  are nested hierarchies.

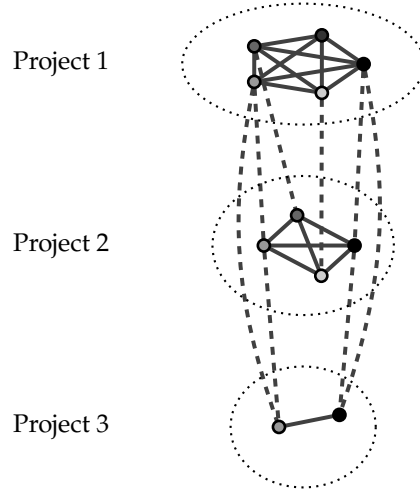


Figure 3: Nested Hierarchy. Vertices are worker-project pairs, and vertices referring to the same worker have the same shade. An edge is present between two vertices sharing the same worker or project coordinate. Solid edges represent effort complementarity, while dashed edges represent effort substitutability.  $G_{\mathcal{F}}$  is necessarily a subgraph of  $G_A$ .

When does the principal underutilize workers? The following is an explicit example of an optimal project assignment in a two-worker, two-project setting. Interestingly, the principal underutilizes a worker at projects that both the worker *and* the principal prefer most given the model primitives.

**Example 1.** There are two workers  $i$  and  $j$  and three projects  $p$ ,  $q$  and  $s$ . The model parameters are as given:

|                                              | Project |      |      |
|----------------------------------------------|---------|------|------|
|                                              | $p$     | $q$  | $s$  |
| Preference $\alpha_i^{(\cdot)}$ , worker $i$ | 0.45    | 0.35 | 0.25 |
| Preference $\alpha_j^{(\cdot)}$ , worker $j$ | 0.60    | 0.45 | 0.20 |
| Principal's weight $\beta^{(\cdot)}$         | 0.75    | 0.50 | 0.40 |

Costs:  $c_i = 0.6, c_j = 0.7, r_i = 0.2, r_j = 0.15$ .  
Synergy:  $g_{ij} = g_{ji} = 0.05$ .  
Valuation:  $V = \sum_p \beta^{(p)} \ln \mathcal{E}^p$ .

Table 1

Every party ranks the projects the same way. However, the optimum does not.

|                                                                           | Project            |                    |              |
|---------------------------------------------------------------------------|--------------------|--------------------|--------------|
|                                                                           | $p$                | $q$                | $s$          |
| <b><math>\hat{\mathbf{e}} = \mathbf{H}^{-1}\boldsymbol{\alpha}</math></b> |                    |                    |              |
| Worker $i$                                                                | 0.667              | 0.379              | 0.069        |
| Worker $j$                                                                | 0.795              | 0.497              | 0.014        |
| Quality $\mathcal{E}^{(\cdot)}$                                           | 1.462              | 0.876              | 0.083        |
| <b>Optimal <math>\mathbf{e}^*</math></b>                                  |                    |                    |              |
| Worker $i$                                                                | 0.396 <sup>†</sup> | 0.289 <sup>†</sup> | 0.191        |
| Worker $j$                                                                | 0.774              | 0.492              | 0.028        |
| Quality $\mathcal{E}^{(\cdot)}$                                           | 1.170              | 0.780              | 0.219        |
| <b>Marginal Utility</b>                                                   |                    |                    |              |
| Worker $i$                                                                | 0.155              | 0.084              | 0            |
| Worker $j$                                                                | 0                  | 0                  | 0            |
| <b>Marginal Value <math>V^{(\cdot)}</math></b>                            | 0.641              | 0.641              | <b>1.827</b> |
| <b>Rank by <math>V^{(\cdot)}</math></b>                                   | <b>2</b>           | <b>2</b>           | <b>1</b>     |

<sup>†</sup> Underutilized: the worker's incentive constraint is slack, so the cap sets his effort.

Table 2

With full utilization, both workers exert a lot of effort at project  $p$  and ignore  $s$ . The principal does the opposite. She suppresses both workers' efforts at project  $p$  so that they can work more on project  $s$ . The principal optimally underutilizes worker  $i$  at project  $p$  and  $q$ , which are the projects that both the principal *and* worker  $i$  prefer most. The underutilization instrument bites exactly where she values effort least on the margin. Project  $s$  carries the smallest weight but the largest marginal value precisely because both workers work so little at it if the principal does not set any caps. Assignments are vacuously nested because  $A^s = A^q = A^p$ , while full utilization sets are strictly nested because  $\mathcal{F}^s = \mathcal{F}^q \supset \mathcal{F}^p$ .<sup>20</sup>

As the previous example hinted, if a worker prefers project  $p$  sufficiently more than project  $s$  and is assigned to both, then the principal must underutilize him at the *former*. The following result formalizes this.

**Proposition 4.** *Assume that worker  $i$  is assigned to both projects  $p$  and  $q$  in the optimum. If  $\alpha_i^p$  is sufficiently larger than  $\alpha_i^q$ , then the principal must underutilize worker  $i$  at project  $p$  and  $\frac{\partial V}{\partial \varepsilon^q} > \frac{\partial V}{\partial \varepsilon^p}$  in the optimum.*

*Proof.* See Appendix A.5. □

This result seems paradoxical at first glance. If worker  $i$  enjoys working on project  $p$  more than  $q$ , then it seems that fully utilizing him on  $p$  is the optimal thing to do. The principal can push him to exert high effort at  $p$ . However, doing so makes a worker exert *too much* effort at  $p$ , violating his incentive constraints at project  $q$ . By Theorem 1, we also know that the marginal value of project  $q$  in the optimum must be greater than that of  $p$  since  $\mathcal{F}^q \supset \mathcal{F}^p$ .

## 4 Designating Specialists and Generalists

Theorem 1 reduces the structure of optimal project assignments to a single ranking by the principal's marginal value and two cutoffs for each worker: one

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<sup>20</sup>Note that the two capped projects share a marginal value is not a knife-edge case. Worker  $i$  is underutilized at both, so  $\lambda_i^p = \lambda_i^q = 0$  and  $\mu_i^p = \mu_i^q = 0$ . Stationarity implies that both projects have the same marginal value  $V^p = V^q$ .

that represents the boundary between assignment and exclusion, and the other between full utilization and underutilization. It stays silent about how these cutoffs are determined, in part because the ordering of project marginal values is endogenous and cannot be deduced from the primitives. In this section, I adopt a parametric approach and vary synergies and costs to see how assignments sort workers into specialists and generalists.

I assume the principal values projects according to a log-linearized Cobb-Douglas function.

$$V(\mathcal{E}^1, \dots, \mathcal{E}^m) = \sum_{p=1}^m \beta^{(p)} \ln(\mathcal{E}^p)$$

Each  $\beta^{(p)}$  weight represents project  $p$ 's relative importance for the principal. Without loss, I order projects by decreasing  $\beta^{(p)}$  and normalize  $\sum_{p=1}^m \beta^{(p)} = 1$ . The Cobb-Douglas objective satisfies Inada conditions for each project, and so the principal must assign at least one worker to each project. I characterize optimal project assignments across three environments: (1) homogeneous workers, (2) heterogeneous multitasking costs, and (3) heterogeneous complementarities.

## 4.1 Homogeneous workers

For all workers  $i$  and projects  $p$ ,  $\alpha_i^p \equiv \alpha$ ,  $r_i \equiv r$ ,  $g \equiv g_{ji}$ . Symmetry permits a closed-form solution.

**Theorem 2.** *There exists a worker-symmetric optimal project assignment. Define the project index*

$$\zeta^* := \max \left\{ \zeta \text{ s.t. } \frac{\sum_{p=1}^{\zeta} \beta^{(p)}}{\beta^{(\zeta)}} - \zeta \leq \frac{c - r - g(n-1)}{r} \right\} \quad 21$$

*The following statements are true of the worker-symmetric optimum.*

1. *It assigns every worker to every project.*

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<sup>21</sup>Under worker homogeneity, [Assumption 1](#) is equivalent to  $c > r + g(n-1)$  as  $c - r - g(n-1)$  is one of the eigenvalues of  $\mathbf{H}$ .



2. The principal fully utilizes every worker at every project  $s \leq \zeta^*$ , and underutilizes every worker at every project  $t > \zeta^*$ .
3. For fully utilized projects  $s \leq \zeta^*$ ,

$$e^{s*} = \frac{\alpha}{c - r - g(n-1) + \zeta^* r} \sum_{p=1}^{\zeta^*} \beta^{(p)}.$$

For underutilized projects  $t > \zeta^*$ ,

$$e^{t*} = \frac{\alpha}{r} \beta^{(t)}.$$

This implies that

$$\mathcal{E}^1 = \dots = \mathcal{E}^{q^*} > \mathcal{E}^{q^*+1} > \dots > \mathcal{E}^m.$$

4.  $\zeta^*$  is non-increasing in  $r$ ,  $g$  and  $n$ , non-decreasing in  $c$  and independent of  $\alpha$ .

*Proof.* See Appendix A.6. □

When workers are identical, the assignment margin shuts down in the worker-symmetric optimum, allowing us to isolate and focus on utilization. Ideally, the principal would like to divide workers' total efforts across projects proportional to their Cobb-Douglas weights; the equation in (3) caps the efforts at high-value projects. Incentive constraints bind, and each worker is contributing as much as they wish; the principal cannot obtain additional effort.  $\zeta^*$  is the index that bisects the project space.<sup>22</sup> Every project that is weakly more valuable than project  $\zeta^*$  receives the same effort. Instead, the principal wields control over the projects with underutilization and sets effort caps that respect the monotonicity of their Cobb-Douglas weights. Notice that heterogeneity in the principal's preferences is what generates underutilization. If for all  $p$  we have  $\beta^{(p)} \equiv \beta$ , then  $\zeta^* = m$ .

The comparative statics of  $\zeta^*$  are characterized by  $\frac{c-r-g(n-1)}{r}$ , which determines the difficulty for a worker to specialize in a particular project. Raising multi-tasking costs  $r$  or collaborative gains  $g$  lowers that fraction, narrowing the space of

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<sup>22</sup>I abuse notation here as  $\zeta$  in Theorem 1 referred to a scalar cutoff used to compare with project marginal values. Here it depicts the cutoff project index.

projects with full utilization. When workers struggle to work on many projects simultaneously, the principal limits their utilization on more projects, pushing them to exert higher effort on more valuable projects. The principal capitalizes on higher gains from collaboration to extract higher quality at more valuable projects at the expense of the less valuable ones. An increase in team size  $n$  strengthens synergies, and so it operates in the same way as an increase in  $g$ . A rise in specialization costs  $c$  makes specialization more expensive, leading the principal to seek a more balanced project assignment. Finally,  $\zeta^*$  does not depend on  $\alpha$ . Intrinsic preference does not affect the relative benefits or costs to spreading or concentrating effort across projects.

## 4.2 Heterogeneous Multitasking Costs

Workers have varying multitasking costs  $r_i$ , but are otherwise identical. The following result demonstrates that multitasking costs are sufficient for determining workers' degree of specialization in an optimal project assignment.

**Theorem 3.** *In an optimal project assignment, the assignment and full utilization cutoffs  $\sigma_i$  and  $\gamma_i$  are sorted such that  $r_i > r_j$  implies  $\sigma_j \leq \sigma_i$  and  $\gamma_j \leq \gamma_i$ .*

*Proof.* See Appendix A.7. □

Every worker works on the most valuable project, whereas the least valuable projects are staffed only by workers who are the best at multitasking. Specialists, those who are assigned to few projects, work together on large project teams, while generalists could potentially be the *only* ones employed at a particular project.

While the extensive margin in assignments and utilization are ordered, the intensive margin is not. Effort levels are sensitive to the model parameters. On the one hand, a high multitasking cost leads the principal to reduce that worker's efforts across multiple projects, holding fixed the set of assigned projects. On the other hand, the principal seeks to assign such a worker to a *smaller* set of assigned projects, freeing up his capacity to work more. The net effect can go either way.

**Example 2.** There are two workers  $i$  and  $j$  and two projects  $p$  and  $q$ . The model parameters are  $\alpha \equiv 1$ ,  $c = 0.6$ ,  $g_{ij} = 0.05$  and multitasking costs  $r_i = 0.15$  and

$r_j = 0.05$ . Worker  $i$  is the specialist and worker  $j$  is the generalist. I increase the principal's  $\beta$  weight on project  $p$  from 6 to 7 while keeping his weight on project  $q$  at 1.

|                                        | $\beta^{(p)} = 6$ |                    |       | $\beta^{(p)} = 7$ |                    |       |
|----------------------------------------|-------------------|--------------------|-------|-------------------|--------------------|-------|
|                                        | $p$               | $q$                | Total | $p$               | $q$                | Total |
| Worker $i$ (specialist, $r_i = 0.15$ ) | 1.662             | 0.577 <sup>†</sup> | 2.239 | 1.724             | 0.329 <sup>†</sup> | 2.053 |
| Worker $j$ (generalist, $r_j = 0.05$ ) | 1.674             | 1.575              | 3.249 | 1.681             | 1.554              | 3.235 |
| Gap $e_j^p - e_i^p$                    | <b>+0.012</b>     |                    |       | <b>−0.044</b>     |                    |       |

<sup>†</sup> Underutilized. Worker  $i$  is capped at project  $q$ ; worker  $j$  is fully utilized everywhere.

Table 3

The intensive margin is not ordered while the extensive margin is unchanged across the two columns. The generalist is fully utilized at both projects, the specialist only at  $p$ , yet the ranking of effort levels at  $p$  reverses.

When  $\beta^{(p)} = 6$ , the generalists works more on  $p$ ; when  $\beta^{(q)} = 7$ , the specialist does. The assignment pattern stays the same. The total effort column reveals why this is true. When the principal is more biased towards project  $p$ , she tightens the specialist's cap at project  $q$ . His total effort falls, dragging his marginal cost down, which frees up his capacity to work more on project  $p$ . Since the generalist is fully utilized at both projects, there is little to no room for the principal to exercise her discretion in directly choosing his efforts. Whether a specialist works more or less than a generalist on a shared project depends on the size of the parameters, and not the inherent structure of the assignment.

### 4.3 Heterogeneous Complementarities

In this final subsection, I assume effort complementarity is heterogeneous across workers, but every other worker parameter remains homogeneous. Cutoffs will be determined by an object I call a worker's *congestion index*.

**Theorem 4.** Define the within-project interaction matrix  $\mathbf{M} := \mathbf{C} - \mathbf{R} - \mathbf{G}$ . Define on a

subset of workers  $S \subseteq N$  the congestion index  $\rho_i$  for  $i \in S$  in the optimum:

$$\rho_i(S) := \frac{r((\mathbf{M}^S)^{-1}\mathbf{\Lambda}^S)_i}{((\mathbf{M}^S)^{-1}\mathbf{1}^S)_i}$$

where  $\mathbf{M}^S$ ,  $\mathbf{\Lambda}^S$  and  $\mathbf{1}^S$  are the submatrices or subvectors with indices restricted to those in  $S$ . Let  $N_0 := N$  and compute recursively for  $k = 1, \dots, n$

$$k := \arg \max_{i \in N_{k-1}} \rho_i(N_{k-1}) \text{ where } N_k := N_{k-1} \setminus \{k\}$$

Relabel worker indices to align with the  $k$  ordering and define  $\rho_k := \rho_k(N_k)$ . The assignment and full utilization cutoffs  $\sigma_i$  and  $\gamma_i$  are sorted such that  $\rho_i > \rho_j$  implies  $\sigma_j \leq \sigma_i$  and  $\gamma_j \leq \gamma_i$ .

*Proof.* See Appendix A.8. □

Which workers are specialists and which are generalists? [Theorem 4](#) boils it down to a single statistic per worker. The principal assigns worker  $i$  to project  $p$  exactly when the project's marginal value exceeds his **congestion index**  $\rho_i$ .  $\rho_i$  is a ratio of two centralities computed using the matrix  $\mathbf{M}$  but with different weights. The diagonals of  $\mathbf{M}$  are  $c_i - r_i$ , which is the change in marginal cost for worker  $i$  at project  $p$  if he increases his effort there, holding his total effort fixed. This can be seen from writing binding incentive constraints for project  $p$  in vector form.

$$\alpha_i^p + \sum_{j \neq i} g_{ij} e_j^p = c_i e_i^p + r_i \sum_{q \neq p} e_i^q \implies \boldsymbol{\alpha}^p = \mathbf{M} \mathbf{e}^p + \mathbf{R}[\mathbf{e}_1 \cdots \mathbf{e}_n] \mathbf{1}_m$$

This is why  $\mathbf{M}$  is deemed the within-project interaction matrix. The denominator of  $\rho_i$  is the canonical Katz-Bonacich (KB) centrality  $((\mathbf{M})^{-1}\mathbf{1})_i$ . It counts the total additional effort that comes from relaxing his incentive constraint at a project, incorporating both the direct effect for worker  $i$  and all of the complementarity effects for other workers. The numerator applies the same interaction matrix to  $\Lambda_i$ , which is the total shadow value of worker  $i$ 's incentive constraints. It measures the opportunity cost incurred by the principal for all other projects when she increases worker  $i$ 's effort towards a specific project, scaled by  $r$ . A high congestion index means it is better to concentrate that worker on only a few projects. In this man-

ner, it is clear why traditional centrality measures do not order the assignment. A worker who has high KB centrality is also one whose incentive constraints are valuable to relax, and therefore has a large  $\Lambda_i$ . From the primitives alone, it is unclear which force dominates.

**Example 3.** In this example, there are 6 workers and 3 projects.

| Worker $i$ | Worker $j$ |       |       |       |       |       |
|------------|------------|-------|-------|-------|-------|-------|
|            | 1          | 2     | 3     | 4     | 5     | 6     |
| 1          | –          | 0.044 | 0.242 | 0.339 | 0.439 | 0.067 |
| 2          | 0.044      | –     | 0.036 | 0.071 | 0.051 | 0.391 |
| 3          | 0.242      | 0.036 | –     | 0.065 | 0.264 | 0.409 |
| 4          | 0.339      | 0.071 | 0.065 | –     | 0.111 | 0.156 |
| 5          | 0.439      | 0.051 | 0.264 | 0.111 | –     | 0.138 |
| 6          | 0.067      | 0.391 | 0.409 | 0.156 | 0.138 | –     |

Synergy matrix  $G$ , six workers.

Table 4

The other model parameters are  $\alpha = 0.5$ ,  $c = 2$ ,  $r = 0.4$ ,  $\beta^{(1)} = 10$ ,  $\beta^{(2)} = 5$  and  $\beta^{(3)} = 1$ . Their KB centralities with respect to  $\mathbf{M}$  and optimal assignments are given by

| Worker $i$ | KB    | Assignments   |
|------------|-------|---------------|
| 1          | 1.739 | $\{1, 2\}$    |
| 2          | 1.712 | $\{1, 2, 3\}$ |
| 3          | 1.685 | $\{1, 2\}$    |
| 4          | 1.664 | $\{1, 2, 3\}$ |
| 5          | 1.398 | $\{1, 2, 3\}$ |
| 6          | 1.244 | $\{1, 2, 3\}$ |

Table 5

The most central worker specializes in the top two projects, rather than being spread across multiple project teams. The worker with the second-highest centrality works on all three projects, while the worker with the fourth-highest centrality only works on projects 1 and 2, breaking the monotonic pattern with respect to Katz-Bonacich Centrality.

Importantly, the relevant network shrinks as we descend the project value ordering. Once a specialist is identified and left off projects below his cutoff  $\rho_i$ , he is no longer a source of complementarity for any worker left at those projects. The congestion index must be recomputed for the remaining workers on the submatrix of  $\mathbf{M}$  *leaving out* worker  $i$ . This is the reason for the recursive updating of  $N_k$  and why the congestion index looks very different than standard centrality measures.

The ordering of workers can be viewed as a sequential key player problem. In [Ballester et al. \(2006\)](#), a planner deletes a player from a network to minimize aggregate economic activity. The player has the highest *intercentrality*, which is his KB centrality but scaled to account for the activity his neighbors lose after he is removed. Here, workers are unassigned from projects when the marginal cost of keeping them on exceeds the marginal benefit. A similar KB index appears but is scaled by an endogenous object  $\Lambda_i$ . The difference is important because when the principal removes a worker from a particular project, his multitasking costs fall, which relaxes his incentive constraints at other projects and boosts both his and his collaborators' efforts. The numerator  $r(\mathbf{M}^{-1}\mathbf{\Lambda})_i$  accounts for this redistribution effect of effort at other projects within the organization. In contrast, there is no redistribution of efforts for the removed player in [Ballester et al. \(2006\)](#).

## 5 Comparative Statics

The comparative statics of an optimal project assignment are governed by which constraints bind according to the assignment and full utilization patterns. I examine the comparative statics across three distinct regimes, ordered by increasing principal involvement.

First, I consider full utilization and assignment, i.e.,  $A^p = \mathcal{F}^p = N$  for every project. Every constraint binds, and the principal's valuation plays no role. There is a full-rank linear system that describes the optimal project assignment, and its derivatives sufficiently pin down the comparative statics. Shocks decompose into a *within-project spillover* and a *network reallocation effect*. The former operates through effort complementarity while the latter captures how multitasking induces workers to redistribute efforts across multiple projects.

Second, I analyze full utilization and partial assignment i.e.  $A^p = \mathcal{F}^p \subset N$  at some project. The linear system remains full-rank, but only on the support of the assignment. Shocks propagate through the organizational network induced by the assignment. Sign predictions in this regime depend on the path a shock travels through the organizational network.

Third, I consider optimal project assignments with underutilization i.e.  $\mathcal{F}^p \subset A^p$  at some project. Slack incentive constraints create a rectangular linear system, and the latter is no longer sufficient for determining efforts. Interpreting binding incentive constraints as a linear budget over workers' efforts, every shock admits a Slutsky decomposition into an income effect and substitution effect. The income effect comprises the aforementioned within-project and network reallocation effects, while the substitution effect accounts for the curvature of the principal's valuation.

## 5.1 Full Utilization and Assignment

Under full utilization and assignment, the optimal project assignment is sufficiently determined by the binding incentive constraints and is given by  $\hat{\mathbf{e}} := \mathbf{H}^{-1}\boldsymbol{\alpha}$ . The following result decomposes any shock into two separate components. There is an immediate effect on a worker's incentive constraint at the shocked project, and a broader network reallocation effect that runs through his total effort, which tightens his incentive constraints at every project. In what ensues, a positive shock induces a *local* benefit to a project but necessarily comes at a *global* cost to the organization.

**Theorem 5.** Define  $\mathbf{M} := \mathbf{C} - \mathbf{R} - \mathbf{G}$  and  $\mathbf{P} := \mathbf{M} + m\mathbf{R}$ . In the optimal project assign-

ment with full utilization and assignment, the following comparative statics hold.

$$\frac{\partial e_i^{p*}}{\partial \alpha_j^q} = \mathbf{M}_{ij}^{-1} \delta_{pq} - \sum_{\ell=1}^n \mathbf{M}_{i\ell}^{-1} r_\ell \mathbf{P}_{\ell j}^{-1}$$

$$\frac{\partial e_i^{p*}}{\partial c_j} = -\mathbf{M}_{ij}^{-1}(e_j^{p*}) + \mathcal{E}_j^* \sum_{\ell=1}^n \mathbf{M}_{i\ell}^{-1} r_\ell \mathbf{P}_{\ell j}^{-1}$$

$$\frac{\partial e_i^{p*}}{\partial r_j} = -\mathbf{M}_{ij}^{-1}(\mathcal{E}_j^* - e_j^{p*}) + (m-1)\mathcal{E}_j^* \sum_{\ell=1}^n \mathbf{M}_{i\ell}^{-1} r_\ell \mathbf{P}_{\ell j}^{-1}$$

$$\frac{\partial e_i^{p*}}{\partial g_{jk}} = \mathbf{M}_{ij}^{-1}(e_k^{p*}) + \mathbf{M}_{ik}^{-1}(e_j^{p*}) - \sum_{\ell=1}^n \mathbf{M}_{i\ell}^{-1} r_\ell (\mathbf{P}_{\ell j}^{-1} \mathcal{E}_\ell^* + \mathbf{P}_{\ell k}^{-1} \mathcal{E}_j^*)$$

where  $\delta_{pq}$  is the Kronecker delta. Moreover  $\frac{\partial e_i^{p*}}{\partial \alpha_j^p} > 0$  for all  $j$  and  $\frac{\partial e_i^{p*}}{\partial \alpha_j^q} < 0$  for all  $q \neq p$ .

*Proof.* See Appendix A.9. □

The inverses of  $\mathbf{M}$  and  $\mathbf{P}$ , which are both nonnegative matrices, play a starring role in the comparative statics. To see the mechanics behind the shock decompositions, I focus on the first comparative static.

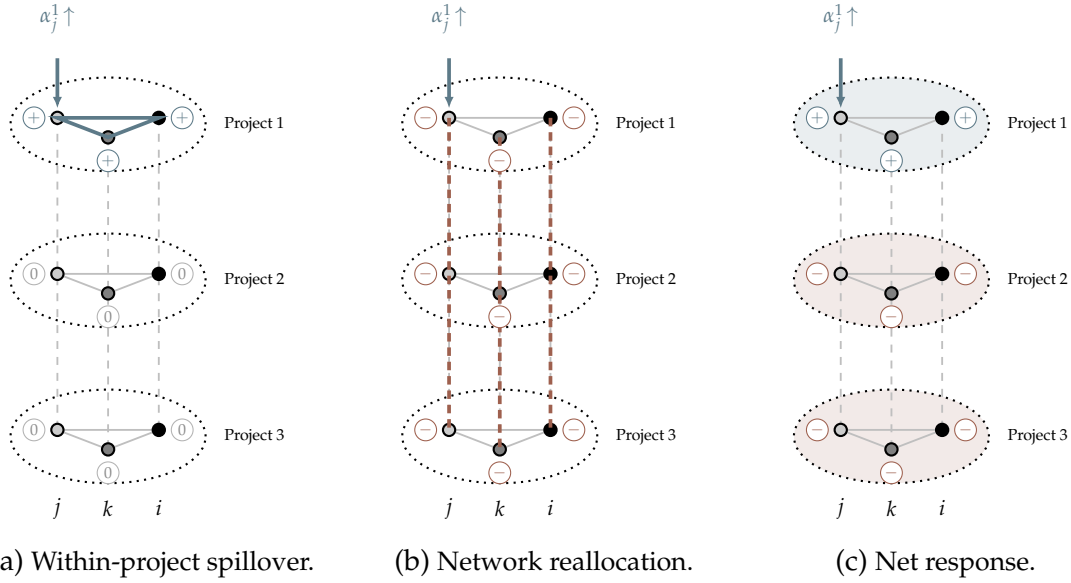
$$\frac{\partial e_i^{p*}}{\partial \alpha_j^q} = \underbrace{\mathbf{M}_{ij}^{-1} \delta_{pq}}_{\text{within-project spillover}} - \underbrace{\sum_{\ell=1}^n \mathbf{M}_{i\ell}^{-1} r_\ell \mathbf{P}_{\ell j}^{-1}}_{\text{network reallocation effect}}$$

Consider a shock to preference  $\alpha_j^p$ . Worker  $j$  is more willing to work on project  $p$ , and complementarity makes everyone else supply more effort at  $p$ . In response, the principal raises every worker's effort cap at project  $p$ . This is captured by the  $(i, j)$ -element of the within-project interaction matrix  $\mathbf{M}^{-1}$ .

The second term represents a separate reallocation channel. The matrix  $\mathbf{P}^{-1}$  satisfies a fundamental equation:  $[\mathbf{e}_1 \cdots \mathbf{e}_n] \mathbf{1}_m = \mathbf{P}^{-1} \sum_{p=1}^m \alpha^p$ . Notice that its elements detail how a worker's *total effort* responds to a preference shock. Now



consider worker  $\ell$ 's response to raising  $\alpha_j^p$ . An increase in worker  $\ell$ 's total effort increases his marginal cost by  $r_\ell$ , making it more expensive for him to exert effort towards *any* project. Consequently, he *lowers* his effort on project  $p$ , effectively reducing the effort complementarities received by worker  $i$  at project  $p$ . The  $\mathbf{M}_{i\ell}^{-1}$  entry captures this effect, and the summation accounts for all workers also working on project  $p$ . The network reallocation effect causes a drop in quality for *every* project not subject to the positive preference shock.



**Figure 4:** Every worker is assigned to and fully utilized at every project. Worker  $j$  receives a preference shock at project 1. The spillover in (a) travels along the solid edges of the shocked ellipse alone. The network reallocation in (b) travels down the dashed edges: higher effort raises each worker's total effort and his marginal cost by  $r_\ell$ , causing him to reduce efforts at every project. Adding the (a) and (b) gives (c): a positive shock local to one project degrades the quality of all others.

Cost shocks behave a bit differently. An increase in  $c_j$  or  $r_j$  necessarily reduces workers' total efforts, but for a project, the two effects push in opposite directions. Lowering total efforts also lowers a worker's marginal cost, inducing him to increase efforts at a project. The sign prediction depends on the exact magnitudes of the efforts in the optimal project assignment.

## 5.2 Full Utilization and Partial Assignment

By leaving some workers off some projects, shock transmission now becomes influenced by the pattern of active project assignments. Shocks still decompose into the two aforementioned effects, but with a couple of crucial differences. Under partial assignment, the primary object of interest is  $\mathbf{M}^p$ , which is the principal submatrix of  $\mathbf{M}$  restricted to workers assigned to  $p$ . Within-project spillovers still survive but are project-dependent and captured by  $(\mathbf{M}^p)^{-1}$ . The network reallocation effect changes as well because each project carries a different  $\mathbf{M}^p$  matrix. The operator that links total efforts to the shock is no longer  $\mathbf{P}$ , but  $\mathcal{P}$  built from the  $\mathbf{M}^p$  matrices. Define  $A_i := \{p \mid e_i^p > 0\}$ .  $\mathcal{P}$  has entries

$$\mathcal{P}_{ij} = \delta_{ij} + r_j \sum_{p \in A_i \cap A_j} (\mathbf{M}^p)^{-1}_{ij}.$$

Its inverse  $\mathcal{P}^{-1}$  fails to be nonnegative, and now workers' total effort responses could move in either direction.

**Theorem 6.** Define the following right-hand vectors  $\sigma^{(\zeta)}$  corresponding to each model parameter  $\zeta$  with entries:

$$\sigma_i^{(\alpha_j^q)} = \begin{cases} (\mathbf{M}^q)^{-1}_{ij} & \text{if } i \in A^q \\ 0 & \text{o.w.} \end{cases} ; \sigma_i^{(c_j)} = - \sum_{p \in A_i \cap A_j} (\mathbf{M}^p)^{-1}_{ij} e_j^{p*}$$

$$\sigma_i^{(r_j)} = \sum_{p \in A_i \cap A_j} (\mathbf{M}^p)^{-1}_{ij} (e_j^{p*} - \mathcal{E}_j^*) ; \sigma_i^{(g_{j\ell})} = \sum_{p \in A_i \cap A_j \cap A_\ell} (\mathbf{M}^p)^{-1}_{ij} e_\ell^{p*} + (\mathbf{M}^p)^{-1}_{i\ell} e_j^{p*}$$

The following equations hold.

$$\frac{\partial e_i^{p*}}{\partial \alpha_j^q} = (\mathbf{M}^p)_{ij}^{-1} \delta_{pq} - \sum_{\ell \in A^p} (\mathbf{M}^p)_{i\ell}^{-1} r_\ell (\mathcal{P}^{-1} \boldsymbol{\sigma}^{(\alpha_j^q)})_\ell$$

$$\frac{\partial e_i^{p*}}{\partial c_j} = -\mathbf{1}_{j \in A^p} (\mathbf{M}^p)_{ij}^{-1} (e_j^{p*}) + \sum_{\ell \in A^p} (\mathbf{M}^p)_{i\ell}^{-1} r_\ell (\mathcal{P}^{-1} \boldsymbol{\sigma}^{(c_j)})_\ell$$

$$\frac{\partial e_i^{p*}}{\partial r_j} = -\mathbf{1}_{j \in A^p} (\mathbf{M}^p)_{ij}^{-1} (\mathcal{E}_j^* - e_j^{p*}) + \sum_{\ell \in A^p} (\mathbf{M}^p)_{i\ell}^{-1} r_\ell (\mathcal{P}^{-1} \boldsymbol{\sigma}^{(r_j)})_\ell$$

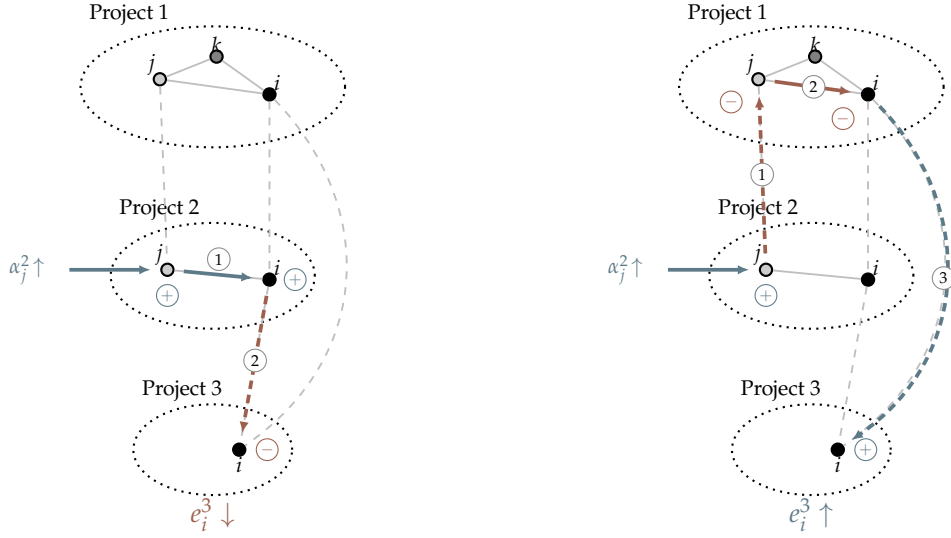
$$\frac{\partial e_i^{p*}}{\partial g_{jk}} = \mathbf{1}_{j,k \in A^p} ((\mathbf{M}^p)_{ij}^{-1} e_k^{p*} + (\mathbf{M}^p)_{ik}^{-1} e_j^{p*}) - \sum_{\ell \in A^p} (\mathbf{M}^p)_{i\ell}^{-1} r_\ell (\mathcal{P}^{-1} \boldsymbol{\sigma}^{(g_{j\ell})})_\ell$$

where  $\mathbf{1}_{j \in A^p}$  is a  $|A^p|$ -dimensional vector that is 1 in the  $j$ -th entry if  $j \in A^p$ , and 0 otherwise.

*Proof.* See Appendix A.10. □

The formulas have a passing resemblance to those of [Theorem 5](#), but the managerial insight is quite different. Under full assignment and utilization, a positive preference shock raised quality at the project subject to the shock and lowered it everywhere else. The principal cannot anticipate the direction of the spillover from such a shock as it depends on the staffing pattern.

**Example 4.** Suppose there are three projects (1, 2, 3) and three workers ( $i, j, k$ ), and the optimum project assignment has the following nested structure: worker  $i$  is assigned to all 3 projects, worker  $j$  is assigned to projects 1 and 2 and worker  $k$  is assigned to project 1.



(a) The direct route  $(j, 2) \rightarrow (i, 2) \rightarrow (i, 3)$ . Complementarity raises worker  $i$ 's effort at project 2; his multitasking cost then pulls him off project 3.

(b) The indirect route,  $(j, 2) \rightarrow (j, 1) \rightarrow (i, 1) \rightarrow (i, 3)$ . Worker  $j$  reduces effort at project 1, worker  $i$  loses complementarity there, and he reduces effort there. This frees up effort for  $i$  to spend at project 3.

Figure 5: Two routes for the same shock under partial assignment. Worker  $i$  is assigned to all three projects, worker  $j$  to projects 1 and 2, and worker  $k$  to project 1 alone.

Figure 5 shows an optimal nested hierarchy. Consider an increase  $\alpha_j^2$ . Worker  $i$  raises his efforts at project 2 due to an increase in complementarity, and cuts back on his efforts at projects 1 and 3. Ambiguity in the direction of the shock comes through another more subtle route. Because worker  $i$  withdraws from project 1, he saves on total effort, which pushes him to work harder on project 3. Whether his total effort increases or decreases due to working harder at project 2 or working less at project 1 depends on the strengths of his complementarity. For example, if  $g_{ij}$  is large, then his gain at project 2 dominates, leading to higher total effort and a contraction at project 3. On the other hand, if  $g_{ij}$  is low, his effort reduction at project 1 dominates, and he is available to work more on project 3. In contrast to the full assignment case, a positive preference shock for one project could *raise* the quality of another project.

### 5.3 Underutilization

When the optimal project assignment features underutilization, binding incentive constraints do not determine optimal effort levels. The principal has discretion in choosing efforts; slack in incentive constraints allows the principal to substitute one worker's effort for another. Now, the curvature of the principal's valuation affects the comparative statics. To recover a Slutsky decomposition, I interpret a binding constraint as a budget for effort.

$$\alpha_i^p = c_i e_i^p + r_i \sum_{q \in P, q \neq p} e_i^q - \sum_{j \in A^p, j \neq i} g_{ij} e_j^p$$

$\alpha_i^p$  is the income of effort available to the principal,  $c_i$  is the own price of worker  $i$ 's effort at project  $p$ ,  $r_i$  is the price of his effort at other projects and  $-g_{ij}$  is a *negative* price on other workers' efforts (which relaxes his incentive constraint). Unlike traditional consumer theory, the principal, who is the consumer here, faces a budget per binding incentive constraint instead of just a single budget.

**Theorem 7.** Let  $\mathcal{F} := \{(i, p) \mid \lambda_i^p > 0\}$ . Define the Jacobian

$$\mathbf{J} = \begin{bmatrix} \nabla_{\mathbf{e}}^2 V & \mathbf{H}_{\mathcal{F}} \\ \mathbf{H}_{\mathcal{F}}^T & \mathbf{0} \end{bmatrix}$$

where  $\mathbf{H}_{\mathcal{F}}$  is the submatrix of  $\mathbf{H}$  with columns restricted to indices in  $\mathcal{F}$ . If  $\nabla_{\mathbf{e}}^2 V$  is strictly negative definite on  $\ker(\mathbf{H}_{\mathcal{F}}^T)$ , then  $\mathbf{J}^{-1}$  exists and its block matrix inverse can be written as

$$\mathbf{J}^{-1} = \begin{bmatrix} (\mathbf{J}^{-1})_{11} & (\mathbf{J}^{-1})_{12} \\ (\mathbf{J}^{-1})_{21} & (\mathbf{J}^{-1})_{22} \end{bmatrix}$$

The following derivatives hold.

$$\frac{\partial \mathbf{e}^*}{\partial \alpha_i^p} = (\mathbf{J}^{-1})_{12} \boldsymbol{\eta}_i^p$$

$$\frac{\partial \mathbf{e}^*}{\partial c_i} = \sum_{p \in P_i} \left( (\mathbf{J}^{-1})_{11} \boldsymbol{\epsilon}_i^p \cdot \lambda_i^{p*} - (\mathbf{J}^{-1})_{12} \boldsymbol{\eta}_i^p \cdot e_i^p \right)$$

$$\frac{\partial \mathbf{e}^*}{\partial r_i} = \sum_{p \in P_i} \left( (\mathbf{J}^{-1})_{11} \boldsymbol{\epsilon}_i^p \cdot \sum_{q \neq p} \lambda_i^q - (\mathbf{J}^{-1})_{12} \boldsymbol{\eta}_i^p \cdot \sum_{q \neq p} e_i^q \right)$$

$$\frac{\partial \mathbf{e}^*}{\partial g_{ij}} = \sum_{p \in P_i \cap P_j} (\mathbf{J}^{-1})_{11} (-\boldsymbol{\epsilon}_i^p \cdot \lambda_j^q - \boldsymbol{\epsilon}_j^p \cdot \lambda_i^p) + (\mathbf{J}^{-1})_{12} (\boldsymbol{\eta}_i^p \cdot e_j^p + \boldsymbol{\eta}_j^p \cdot e_i^p)$$

where  $\boldsymbol{\epsilon}_i^p$  and  $\boldsymbol{\eta}_i^p$  are  $mn \times 1$  and  $|\mathcal{F}| \times 1$  unit vectors that are 1 at the  $(i, p)$  index and 0 everywhere else.

*Proof.* See Appendix [A.11](#). □

The theorem applies the Implicit Function Theorem, and  $\mathbf{J}_{11}^{-1}$  is the resulting Slutsky matrix, which describes the substitution effects. It is symmetric, negative semidefinite and satisfies  $\mathbf{H}_{\mathcal{F}}^\top \mathbf{J}_{11}^{-1} = \mathbf{0}$ . The last property ensures substitution is available and encodes the relative price effects on the optimal reallocation of efforts holding incomes  $\alpha$  fixed. Geometrically, the incentive constraints form a polytope, and underutilization means optimal efforts sit on a face of the polytope. This is exactly why the substitution effect was nonexistent in [Theorem 5](#) and [Theorem 6](#); if every incentive constraint binds, then the optimal project assignment resides at a vertex of the polytope. In other words, the preceding theorems only exhibit income effects.

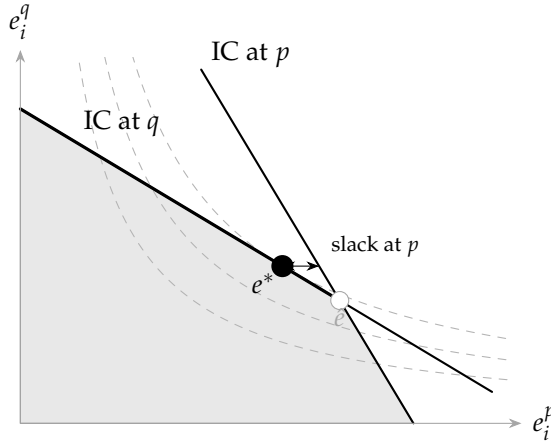


Figure 6: Underutilization at  $\mathbf{e}^*$  for one worker and two projects.

The formula for  $\frac{\partial \mathbf{e}}{\partial \alpha_i^p}$  captures the income effect, which is embedded in the other comparative statics. Take an increase in specialization cost for worker  $i$ :

$$\frac{\partial \mathbf{e}^*}{\partial c_i} = \sum_{p \in P_i} \left( (\mathbf{J}^{-1})_{11} \epsilon_i^p \cdot \lambda_i^p - (\mathbf{J}^{-1})_{12} \eta_i^p \cdot e_i^p \right) = \sum_{p \in P_i} \left( (\mathbf{J}^{-1})_{11} \epsilon_i^p \cdot \lambda_i^p - \frac{\partial \mathbf{e}}{\partial \alpha_i^p} \cdot e_i^p \right)$$

A rise in  $c_i$  makes  $\mathbf{e}$  more expensive overall to implement. With a fixed income  $\alpha$ , the principal scales back efforts, captured by  $(\mathbf{J}^{-1})_{12}$ . In contrast, it also makes worker  $i$ 's efforts relatively more costly. The principal substitutes away from worker  $i$ 's effort in favor of other workers' efforts, captured by the Slutsky matrix  $(\mathbf{J}^{-1})_{11}$ . Finally, I draw a major distinction between the analysis here and conventional consumer theory. Traditionally, each price corresponds to only one good. The prices of the goods, which are efforts, affect multiple budgets simultaneously, represented by the aggregation of income and substitution effects in the formulas.

## 6 Discussion

### 6.1 Asymmetric Complementarities

I imposed symmetric complementarities early on to write the workers' effort game as an exact potential game, which buys us uniqueness of equilibrium. Beyond that, complementarities can be asymmetric, and all of the main results still hold.

Worker interactions must be complements and not substitutes within a project. If  $g_{ij} < 0$ , then the argument that shadow values of incentive constraints move in the same direction when project marginal values change no longer holds. Formally, the within-project spillover matrix  $\mathbf{M} = \mathbf{C} - \mathbf{R} - \mathbf{G}$  is no longer an  $M$ -matrix because it has negative off-diagonals, and so  $\mathbf{M}^{-1}$  may have negative entries. The nested hierarchy structure no longer holds.

### 6.2 Effort Budgets

As the leading examples in the introduction suggested, firms and organizations typically allocate the *share* of effort workers have across different projects. That is, there is a ceiling to the effort cap the principal can set. This restriction places an additional constraint in the principal's convex optimization problem where for all workers  $i$ ,  $\sum_{p=1}^m e_i^p \leq T_i$  where  $T_i$  is an exogenously given hard limit. The main nested hierarchy result still holds. The stationarity condition in this extension is given by

$$(c_i - r_i)\lambda_i^p - \mu_i^p = V^p + \sum_{j \neq i} g_{ij}\lambda_j^p - r_i\Lambda_i - v_i$$

where  $v_i$  is the multiplier on the external budget constraint. Notice that it is worker-specific, and so we can analyze each project separately *holding*  $r_i\Lambda_i$  and  $v_i$  fixed. Once this observation is made, the rest of the proof of the result goes through.

What changes are the results governing where the cutoffs lie in the parametric Cobb-Douglas setting. The budget is a second sorting force that competes with multitasking costs, and whether workers are designated specialists or generalists depends on how lax their effort budget constraint is.



### 6.3 Effort Caps versus Contracts

By studying optimal project assignments via effort caps in isolation, I prohibit the principal from contracting on effort. A natural question is whether she could achieve the same if transfers were allowed. Effort caps are not a convenient substitute for contracts the principal could write, and the outcomes implementable by either are generally disjoint. I show that effort caps implement effort profiles that linear contracts implement with weakly negative wages, and those with underutilization require strictly negative wages. Limited liability would disallow such a contract in almost every setting.

To see this concretely, assume that in place of effort caps, the principal chooses a linear contract for each worker  $i$

$$w_i(\mathbf{e}_i) = \sum_{p=1}^m b_i^p e_i^p \text{ where } b_i^p \in \mathbb{R}.$$

Upon choosing contracts, workers play the effort game and receive augmented payoffs  $U_i(\mathbf{e}_i, \mathbf{e}_{-i}) + w_i(\mathbf{e}_i)$ .<sup>23</sup>

**Proposition 5.** *Wage rates  $\mathbf{b} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$  implement  $\mathbf{e} \geq \mathbf{0}$  as the unique Nash equilibrium of the effort game with transfers if and only if*

$$b_i^p = -(\alpha_i^p + \sum_{j \neq i} g_{ij} e_j^p - c_i e_i^p - r_i \sum_{q \neq p} e_i^q) \text{ if } e_i^p > 0 \quad b_i^p \leq -(\alpha_i^p + \sum_{j \neq i} g_{ij} e_j^p - c_i e_i^p - r_i \sum_{q \neq p} e_i^q) \text{ if } e_i^p = 0$$

*Proof.* See [Section A.12](#). □

To implement an effort profile, the required rate is the negative of the worker's marginal utility at the target. To underutilize a worker and leave his incentive constraint slack, the principal must fine him for working. Effort caps and *nonnegative* wages are different instruments that move effort in the opposite direction. A cap holds a worker back on a project he would like to contribute more to, while a positive wage pushes him to do so. Instead, contracts can achieve effort profiles where worker marginal utilities are negative, something effort caps cannot accomplish.

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<sup>23</sup>I leave out a fixed payment because it does not affect workers' incentives to exert effort.

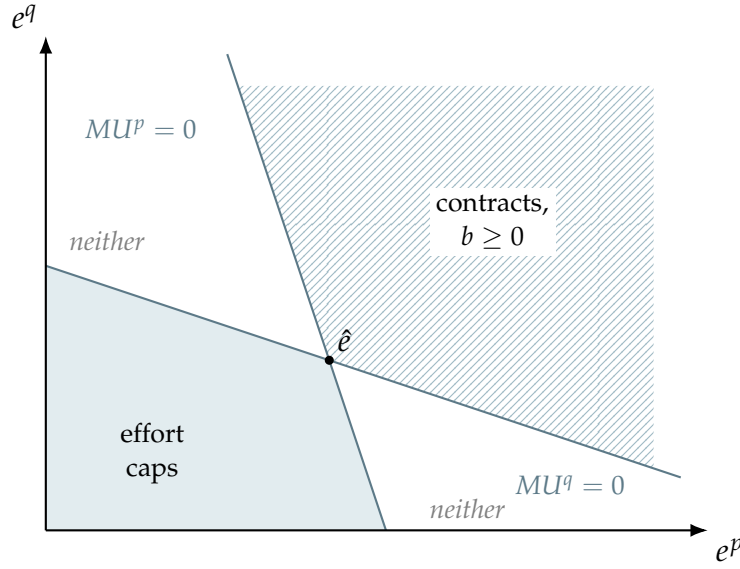


Figure 7: Effort profiles implementable by caps and by contracts, for one worker and two projects. The lines are the worker’s incentive constraints. Effort caps implement exactly the profiles at which marginal utility is nonnegative at both projects (shaded), by setting  $\bar{e} = e$ . Nonnegative linear contracts implement exactly those at which marginal utility is nonpositive at both (hatched, unbounded). The two sets meet only at the unconstrained equilibrium  $\hat{e}$ , where every marginal utility vanishes. In the two unshaded regions neither instrument alone suffices.

## 6.4 Worker Welfare

Effort caps redirect effort towards projects the principal values more on the margin, and the numerics from [Example 1](#) allow us to compute what it costs the workers explicitly.

| Worker | $U(\hat{e})$ | $U(e^*)$ | $\Delta$ |
|--------|--------------|----------|----------|
| $i$    | 0.243        | 0.218    | -10.4%   |
| $j$    | 0.370        | 0.357    | -3.4%    |

Both workers lose utility moving from the unconstrained equilibrium to the principal’s optimum. Because the principal’s objective is to maximize her joint valuation of project quality, not worker utilities, there is no reason for the use of effort caps to be socially desirable. In fact, this example demonstrates how it is detrimental to all workers within an organization. Identifying which workers lose more is a relevant welfare implication that lies beyond the scope of this paper.

## 7 Concluding Remarks

Organizations that do not pay for effort on a day-to-day basis regularly choose how much they want implemented. Effort caps are not a substitute for incentives and deserve their own analysis. It is essentially two instruments in one: it selects which workers collaborate with whom, and it dictates the breadth of what a worker works on in the organization. Optimally assigning workers to projects by setting effort caps turns out to be far more simple than the richness of the model suggests. A single endogenous ranking of projects sorts every worker's schedule into three consecutive intervals—fully utilized, assigned but underutilized, unassigned ([Theorem 1](#)). Workers tend to be underutilized at projects they most prefer. Because the assignment selects the network of interactions, an organization's response to shocks is implicitly designed by the principal. A shock favorable to one project could end up being bad news for other projects in the organization, depending on the pattern of assignment and utilization in the optimum.

There are many promising extensions for future research. Because this paper studies assignment in isolation, a natural next step is to study the joint design of assignments and incentives, especially given that they affect effort in opposite ways. Project assignments could also be studied in tandem with career concerns, as success on certain projects could advance a worker's career more than others. Lastly, [Assumption 1](#) guarantees a unique equilibrium and nullified equilibrium selection concerns. Relaxing it allows us to study which project assignments remain robust when the worst equilibrium is played. I hope that this work inspires future research in studying staffing decisions and project assignment through the lens of a network design problem.

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## A Omitted Proofs

### A.1 Proof of Proposition 1

( $\Rightarrow$ ). Assume that  $\mathbf{e}$  is a Nash equilibrium given  $\bar{\mathbf{e}}$ . If  $e_i^p < \bar{e}_i^p$ , then

$$\frac{\partial U_i}{\partial e_i^p} = 0 \implies \alpha_i^p + \sum_{j \neq i} g_{ij} e_j^p = c_i e_i^p + r_i \sum_{q \neq p} e_i^q$$

If  $e_i^p = \bar{e}_i^p$ , and worker  $i$ 's marginal utility at project  $p$  is negative, then he profits from reducing his efforts, contradicting that he is in a Nash equilibrium. Because worker utility is concave in own efforts, the first-order conditions are necessary and sufficient.

( $\Leftarrow$ ). I show that  $\mathbf{e}$  with  $\alpha_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q$  whenever  $e_i^p > 0$  is an equilibrium with  $\bar{\mathbf{e}} = \mathbf{e}$ . Because worker utilities are concave in own efforts, a weakly positive marginal utility at any boundary point means that reducing effort cannot be a profitable deviation. Therefore  $e_i^p$  is a best response for worker  $i$  at project  $p$  given effort caps  $\bar{\mathbf{e}}$ .  $\square$

### A.2 Proof of Proposition 2

Assume that  $r_i \tilde{\mathbf{E}}_i \leq \min_p \alpha_i^p$ . Because  $\mathbf{P}$  has a nonnegative diagonal and has negative off-diagonals and has positive eigenvalues by Assumption 1, it is an  $M$ -matrix and  $\tilde{\mathbf{E}} > 0$ . By construction, it satisfies

$$(c_i + r_i(m-1))\tilde{\mathbf{E}}_i = \sum_{p=1}^m \alpha_i^p + \sum_{j \neq i} g_{ij} \tilde{\mathbf{E}}_j$$

Let  $\mathcal{A}_i^{(k)}$  be the set of the  $k$  largest entries of  $\{\alpha_i^p\}_{p=1}^m$ . Then subtract to obtain

$$(c_i + r_i(m-1))\tilde{\mathbf{E}}_i - \sum_{p \in \mathcal{A}_i^{(k)}} \alpha_i^p - \sum_{j \neq i} g_{ij} \tilde{\mathbf{E}}_j = \sum_{p \notin \mathcal{A}_i^{(k)}} \alpha_i^p \geq (m-k) \min_p \alpha_i^p \geq (m-k) r_i \tilde{\mathbf{E}}_i$$

Rearrange to get

$$\begin{aligned} (m-k)r_i\tilde{\mathbf{E}}_i &\leq (c_i + r_i(m-1))\tilde{\mathbf{E}}_i - \sum_{p \in \mathcal{A}_i^{(k)}} \alpha_i^p - \sum_{j \neq i} g_{ij}\tilde{\mathbf{E}}_j \\ \implies (c_i + r_i(k-1))\tilde{\mathbf{E}}_i &\geq \sum_{p \in \mathcal{A}_i^{(k)}} \alpha_i^p + \sum_{j \neq i} g_{ij}\tilde{\mathbf{E}}_j (*) \end{aligned}$$

Now I show that for any implementable  $\mathbf{e}$ , the total efforts  $\mathcal{E}_i := \sum_{p=1}^m e_i^p$  are bounded from above by  $\tilde{\mathbf{E}}_i$ . Summing worker  $i$ 's incentive constraints over projects  $p \in P_i$  gives

$$\alpha_i^p + \sum_{j \neq i} g_{ij}e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q \implies \sum_{p \in P_i} \alpha_i^p + \sum_{j \neq i} g_{ij} \sum_{p \in P_i} e_j^p \geq \sum_{p \in P_i} (c_i e_i^p + r_i \sum_{q \neq p} e_i^q)$$

Write  $\mathcal{E}_i := \sum_{p=1}^m e_i^p$ . The cost side simplifies to

$$\sum_{p \in P_i} (c_i e_i^p + r_i \sum_{q \neq p} e_i^q) = (c_i + r_i(|P_i| - 1))\mathcal{E}_i$$

Now suppose that  $\zeta := \max_i (\mathcal{E}_i / \tilde{\mathbf{E}}_i) > 1$ . Then multiply and divide by  $\tilde{\mathbf{E}}_i$  both the left and right side to get

$$\zeta(c_i + r_i(|P_i| - 1))\tilde{\mathbf{E}}_i \leq \sum_{p \in P_i} \alpha_i^p + \zeta \sum_{j \neq i} g_{ij}\tilde{\mathbf{E}}_j = \sum_{p \in P_i} \alpha_i^p + \sum_{j \neq i} g_{ij}\tilde{\mathbf{E}}_j + (\zeta - 1) \sum_{j \neq i} g_{ij}\tilde{\mathbf{E}}_j$$

Applying the (\*) relation, we have

$$\sum_{p \in P_i} \alpha_i^p + \sum_{j \neq i} g_{ij}\tilde{\mathbf{E}}_j + (\zeta - 1) \sum_{j \neq i} g_{ij}\tilde{\mathbf{E}}_j \leq (c_i + r_i(|P_i| - 1))\tilde{\mathbf{E}}_i + (\zeta - 1) \sum_{j \neq i} g_{ij}\tilde{\mathbf{E}}_j$$

Chaining the inequality from the two lines above:

$$(\zeta - 1)(c_i + r_i(|P_i| - 1))\tilde{\mathbf{E}}_i \leq (\zeta - 1) \sum_{j \neq i} g_{ij}\tilde{\mathbf{E}}_j \implies (c_i + r_i(|P_i| - 1))\tilde{\mathbf{E}}_i \leq \sum_{j \neq i} g_{ij}\tilde{\mathbf{E}}_j$$

This implies that  $\sum_{p \in \mathcal{A}_i^{(k)}} \alpha_i^p \leq 0$ , which is a direct contradiction that  $\alpha_i^p > 0$  for all  $i$  and  $p$ .

Now assume  $\mathbf{e}$  is implementable with  $e_i^p = 0$ . Then  $\sum_{q \neq p} e_i^q = \mathcal{E}_i \leq \tilde{\mathbf{E}}_i$  and so

$$\alpha_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq \min_p \alpha_i^p \geq r_i \tilde{\mathbf{E}}_i \geq r_i \sum_{q \neq p} e_i^q$$

Because worker  $i$ 's incentive constraint at  $p$  is always satisfied given zero effort  $e_i^p$ , every implementable profile must therefore satisfy every incentive constraint and the feasible set of the mixed-integer nonlinear program in [Section C](#) coincides with that of the convex program. Notice that with homogeneous workers,  $\mathbf{P} = (c + (m-1)r + g)\mathbf{I}_n - g\mathbf{1}_n\mathbf{1}_n^\top$  so  $\tilde{\mathbf{E}}_i = \frac{m\alpha}{c + (m-1)r - g(n-1)}$ . Therefore, the following conditions are equivalent

$$r\tilde{\mathbf{E}}_i \leq \alpha \iff rm \leq c + (m-1)r - g(n-1) \iff r + g(n-1) \leq c$$

Under worker homogeneity, [Assumption 1](#) is equivalent to  $c > r + g(n-1)$  as  $c - r - g(n-1)$  is one of the eigenvalues of  $\mathbf{H}$ , which must be positive.  $\square$

### A.3 Proof of [Proposition 3](#)

The KKT condition of the principal's problem, which is necessary and sufficient for optimality as all constraints are linear in efforts and the objective is concave, in vector form, is:

$$\nabla_{\mathbf{e}} V(\mathcal{E}^1, \dots, \mathcal{E}^m) + ((\mathbf{G} \otimes \mathbf{I}_m) - (\mathbf{C} \otimes \mathbf{I}_m + \mathbf{R} \otimes (\mathbf{1}_m \mathbf{1}_m^\top - \mathbf{I}_m)))\boldsymbol{\lambda} + \boldsymbol{\mu} = \mathbf{0}$$

where  $\boldsymbol{\lambda} \geq \mathbf{0}$  and  $\boldsymbol{\mu} \geq \mathbf{0}$  are the vectors of Lagrange multipliers for the incentive constraints and nonnegativity conditions. This is equivalent to  $\nabla_{\mathbf{e}} V(\mathcal{E}^1, \dots, \mathcal{E}^m) = \mathbf{H}\boldsymbol{\lambda} - \boldsymbol{\mu}$ . If  $(i, p) \in A^p$ , then  $\frac{\partial V}{\partial \mathcal{E}^p} = (\mathbf{H}\boldsymbol{\lambda})_i^p$ . If  $(i, p) \notin A^p$ , then  $\frac{\partial V}{\partial \mathcal{E}^p} \leq (\mathbf{H}\boldsymbol{\lambda})_i^p$ . If every worker is assigned and fully utilized at every project, then  $\boldsymbol{\mu} = \mathbf{0}$ , and  $\nabla_{\mathbf{e}} V(\mathcal{E}^1, \dots, \mathcal{E}^m) = \mathbf{H}\boldsymbol{\lambda}$ , which means  $\nabla_{\mathbf{e}} V$  lies in the convex cone generated by the columns of the matrix  $\mathbf{H}$ .  $\square$

## A.4 Proof of Theorem 1

The stationarity condition for the Lagrangian of the principal's optimization problem with respect to  $e_i^p$  is

$$\frac{\partial V}{\partial \mathcal{E}^p} + \sum_{j \neq i} g_{ij} \lambda_j^p - c_i \lambda_i^p - r_i \sum_{q \neq p} \lambda_i^q + \mu_i^p = 0$$

Written in terms of worker  $i$ 's sum of multipliers on the incentive constraints  $\Lambda_i := \sum_{p=1}^m \lambda_i^p$  is:

$$\frac{\partial V}{\partial \mathcal{E}^p} + \sum_{j \neq i} g_{ij} \lambda_j^p + (r_i - c_i) \lambda_i^p - r_i \Lambda_i + \mu_i^p = 0$$

Our goal is to understand the structure of  $\lambda_i^p$  and  $\mu_i^p$  in the optimum. I first observe that the strict slackness condition where  $e_i^p = 0$  implies  $\lambda_i^p = 0$  is a generic condition of the optimum. Appendix B details the full argument why it is so. I stress that the converse is *not* generically true, as the principal underutilizes workers at projects she assigns them to. Using this, I exploit a linear complementarity relationship between  $\lambda_i^p$  and  $\mu_i^p$ .

1. If  $\lambda_i^p > 0$ , then  $e_i^p > 0$  and  $\mu_i^p = 0$ .
2. If  $\mu_i^p > 0$ , then  $e_i^p = 0$  and  $\lambda_i^p = 0$ .

Rearrange the stationarity condition for  $e_i^p$ :

$$\mu_i^p = (c_i - r_i) \lambda_i^p - \sum_{j \neq i} g_{ij} \lambda_j^p + r_i \Lambda_i - \frac{\partial V}{\partial \mathcal{E}^p} \implies \boldsymbol{\mu}^p = \mathbf{M} \boldsymbol{\lambda}^p + \mathbf{R} \boldsymbol{\Lambda} - \left( \frac{\partial V}{\partial \mathcal{E}^p} \right) \mathbf{1}_n$$

where  $\mathbf{M} := \mathbf{C} - \mathbf{R} - \mathbf{G}$  and  $\boldsymbol{\Lambda} = [\Lambda_1, \dots, \Lambda_n]^\top$ . For shorthand notation, I write  $\mathbf{w}^p = \mathbf{R} \boldsymbol{\Lambda} - \left( \frac{\partial V}{\partial \mathcal{E}^p} \right) \mathbf{1}_n$ .

So far, for each project  $p$ , we have effectively formulated a linear complementarity problem in the Lagrange multipliers  $\boldsymbol{\mu}^p$  and  $\boldsymbol{\lambda}^p$ , which I label LCP( $\mathbf{M}, \mathbf{w}^p$ ). Since  $\mathbf{H}$  is positive definite by Assumption 1,  $\mathbf{M}$  is an  $M$ -matrix. The following are two results established by linear complementarity theory.

1. Because  $\mathbf{M}$  is positive definite, LCP( $\mathbf{M}, \mathbf{w}^p$ ) has a unique solution for each right-hand vector  $\mathbf{w}^p$  (Theorem 3.1.6 of Cottle et al. (2009)).

2. Because  $\mathbf{M}$  is a Z-matrix, the unique solution  $\lambda^*(\mathbf{w}^p)$  to  $\text{LCP}(\mathbf{M}, \mathbf{w}^p)$  is the least element of the feasible set (Theorem 3.11.6 of [Cottle et al. \(2009\)](#)).

These two facts help us establish the following monotonicity result.

**Lemma 1.** *If  $\mathbf{w} \leq \mathbf{w}'$  (component-wise), then  $\lambda^*(\mathbf{w}) \geq \lambda^*(\mathbf{w}')$  and  $\mu^*(\mathbf{w}) \leq \mu^*(\mathbf{w}')$ .*

*Proof.* Let  $\mathbf{w} \leq \mathbf{w}'$  (component-wise). Then  $\mathbf{M}\lambda + \mathbf{w}' \geq \mathbf{M}\lambda + \mathbf{w}$ . This implies that  $\{\lambda \mid \mathbf{M}\lambda + \mathbf{w} \geq \mathbf{0}\} \subseteq \{\lambda \mid \mathbf{M}\lambda + \mathbf{w}' \geq \mathbf{0}\}$ . By the least-element property, the solution  $\lambda^*$  must satisfy:

$$\lambda^*(\mathbf{w}') = \min_{\lambda} \{\lambda \mid \mathbf{M}\lambda + \mathbf{w}' \geq \mathbf{0}\} \leq \min_{\lambda} \{\lambda \mid \mathbf{M}\lambda + \mathbf{w} \geq \mathbf{0}\} = \lambda^*(\mathbf{w})$$

The above establishes reverse monotonicity of  $\lambda^*$ . Now I show the monotonicity of  $\mu^*$ . Fix some index  $i$ .

Case #1:  $\lambda^*(\mathbf{w})_i > 0$ .

By complementarity,  $\mu^*(\mathbf{w})_i = 0$ . Since  $\mu^*(\mathbf{w}') \geq \mathbf{0} \implies \mu^*(\mathbf{w}')_i \geq \mu^*(\mathbf{w})_i$ .

Case #2:  $\lambda^*(\mathbf{w})_i = 0$ . Because  $\lambda^*(\mathbf{w}') \leq \lambda^*(\mathbf{w})$ , we have that  $\lambda^*(\mathbf{w}')_i = 0$  as well. Then we can write:

$$\mu^*(\mathbf{w})_i - \mu^*(\mathbf{w}')_i = (\mathbf{M}(\lambda^*(\mathbf{w}) - \lambda^*(\mathbf{w}')))_i + w_i - w'_i$$

Now we use the fact that  $\mathbf{M}$  is a Z-matrix. Since  $\lambda^*(\mathbf{w})_i - \lambda^*(\mathbf{w}')_i = 0$  and  $\lambda^*(\mathbf{w})_j - \lambda^*(\mathbf{w}')_j \geq 0$  for all  $j \neq i$ ,

$$(\mathbf{M}(\lambda^*(\mathbf{w}) - \lambda^*(\mathbf{w}')))_i = \sum_{j \neq i} \mathbf{M}_{ij}(\lambda^*(\mathbf{w})_j - \lambda^*(\mathbf{w}')_j) \leq 0$$

Since  $w'_i \geq w_i$ , we can conclude that  $\mu^*(\mathbf{w})_i - \mu^*(\mathbf{w}')_i \leq 0$ , completing the proof.  $\square$

Returning to the main proof, assume that projects  $p$  and  $q$  are such that  $\frac{\partial V}{\partial \mathcal{E}^p} > \frac{\partial V}{\partial \mathcal{E}^q}$ .

$$\mathbf{w}^p = \mathbf{R}\Lambda - \left( \frac{\partial V}{\partial \mathcal{E}^p} \right) \mathbf{1}_n \leq \mathbf{R}\Lambda - \left( \frac{\partial V}{\partial \mathcal{E}^q} \right) \mathbf{1}_n = \mathbf{w}^q$$

Applying the monotonicity properties of  $\lambda^*$  gives  $\lambda^{p*} \geq \lambda^{q*}$  and  $\mu^{p*} \leq \mu^{q*}$ . Therefore, there exist cutoffs  $\sigma_i$  and  $\gamma_i$  such that

$$\frac{\partial V}{\partial \mathcal{E}^p} \geq \sigma_i \iff \mu_i^p = 0 \iff i \in A^p \qquad \frac{\partial V}{\partial \mathcal{E}^p} \geq \gamma_i \iff \lambda_i^p > 0 \iff i \in \mathcal{F}^p$$

$\gamma_i \geq \sigma_i$ , otherwise there exists project  $p$  such that  $i \in \mathcal{F}^p$  but  $i \notin A^p$ , which is ruled out by [Assumption 2](#). For the last statement, order projects by the magnitude of  $\frac{\partial V}{\partial \mathcal{E}^p}$  in the optimum such that

$$\frac{\partial V}{\partial \mathcal{E}^1} \geq \dots \geq \frac{\partial V}{\partial \mathcal{E}^m}$$

If worker  $i$  is fully utilized at project  $q$  i.e.  $\lambda_i^q > 0$ , then  $\lambda_i^p > 0$ . This implies the set of workers fully utilized at project  $q$  is a subset of those fully utilized at project  $p$ . We arrive at the desired nested structure

$$\mathcal{F}^1 \supseteq \mathcal{F}^2 \supseteq \dots \supseteq \mathcal{F}^m.$$

where  $\mathcal{F}^p = \{i \mid \lambda_i^p > 0\}$ .

If  $e_i^{q*} > 0$ , then  $\mu_i^q = 0$  and so  $\mu_i^p = 0$ . This implies the workers assigned positive effort at project  $q$  are also assigned positive effort at project  $p$  i.e.  $A^q \subseteq A^p$  for all  $p < q$  where  $A^q = \{i \mid e_i^{p*} > 0\}$ , ignoring the knife-edge case where  $e_i^p = 0$  and  $\mu_i^p = 0$ . We arrive at the desired nested structure

$$A^1 \supseteq A^2 \supseteq \dots \supseteq A^m.$$

□

## A.5 Proof of [Proposition 4](#)

Assume that worker  $i$  is fully utilized at project  $p$  in  $\mathbf{e}^*$ . Then the following equation holds:

$$\alpha_i^p + \sum_{j \in A^p \setminus i} g_{ji} e_j^{p*} = c_i e_i^{p*} + \sum_{q \in P_i} r_i e_i^{q*}$$

Solving for  $e_i^{p*}$  yields:

$$e_i^{p*} = \frac{\alpha_i^p + \sum_{j \in A^p \setminus i} g_{ji} e_j^{p*} - \sum_{s \in P_i \setminus p} r_i e_i^{s*}}{c_i}$$

The incentive constraint for  $e_i^{q*}$  is:

$$\alpha_i^q + \sum_{j \in A^q \setminus i} g_{ji} e_j^{q*} \geq c_i e_i^{q*} + \sum_{s \in P_i \setminus q} r_i e_i^{s*}$$

Substituting in the formula for  $e_i^{p*}$  gives:

$$\alpha_i^q + \sum_{j \in A^q \setminus i} g_{ji} e_j^{q*} \geq c_i e_i^{q*} + r_i \left( \frac{\alpha_i^p + \sum_{j \in A^p \setminus i} g_{ji} e_j^{p*} - \sum_{s \in P_i \setminus p} r_i e_i^{s*}}{c_i} \right) + \sum_{s \in P_i \setminus p, q} r_i e_i^{s*}$$

Multiplying both sides by  $c_i$ :

$$c_i \left( \alpha_i^q + \sum_{j \in A^q \setminus i} g_{ji} e_j^{q*} \right) \geq c_i^2 e_i^{q*} + r_i \left( \alpha_i^p + \sum_{j \in A^p \setminus i} g_{ji} e_j^{p*} - \sum_{s \in P_i \setminus p} r_i e_i^{s*} \right) + c_i \sum_{s \in P_i \setminus p, q} r_i e_i^{s*}$$

Rearranging terms, the following inequality is necessary for the above to hold because  $r_i \sum_{j \in A^p \setminus i} g_{ji} e_j^{p*} \geq 0$ :

$$c_i \left( \alpha_i^q + \sum_{j \in A^q \setminus i} g_{ji} e_j^{q*} \right) \geq (c_i^2 - r_i^2) e_i^{q*} + r_i \left( \alpha_i^p - \sum_{s \in P_i \setminus p, q} r_i e_i^{s*} \right) + c_i \sum_{s \in P_i \setminus p, q} r_i e_i^{s*}$$

Proceeding onwards, the following inequality is necessary for the above to hold because  $c_i^2 - r_i^2 > 0$ :

$$c_i \left( \alpha_i^q + \sum_{j \in A^q \setminus i} g_{ji} e_j^{q*} \right) \geq r_i \alpha_i^p + \sum_{s \in P_i \setminus p, q} (c_i r_i - r_i^2) e_i^{s*}$$

Since  $c_i r_i - r_i^2 > 0$ , the following inequality is necessary for the above to hold:

$$c_i \left( \alpha_i^q + \mathbf{G}_{i,\cdot}^q \cdot (\mathbf{C}^q - \mathbf{G}^q)^{-1} \boldsymbol{\alpha}^q \right) \geq r_i \alpha_i^p$$

where  $\mathbf{C}^q$  is submatrix of  $\mathbf{C}$  restricted to workers assigned positive effort at project  $q$ , and  $\mathbf{G}^q$  is the submatrix of  $\mathbf{G}$  restricted to workers assigned positive effort at project  $q$ . If  $\alpha_i^p$  is sufficiently large, i.e.

$$\alpha_i^p > \frac{c_i \left( \alpha_i^q + \mathbf{G}_{i,\cdot}^q \cdot (\mathbf{C}^q - \mathbf{G}^q)^{-1} \boldsymbol{\alpha}^q \right)}{r_i}$$

then worker  $i$  cannot be fully utilized at project  $p$  in  $\mathbf{e}^*$ . Since  $\mathcal{F}^q \supseteq \mathcal{F}^p$ , it must be that  $\frac{\partial V}{\partial \mathcal{E}^q} \geq \frac{\partial V}{\partial \mathcal{E}^p}$ .  $\square$

## A.6 Proof of Theorem 2

I first show that there always exists a worker-symmetric optimal project assignment. Note that the principal's objective is strictly concave over  $(\mathcal{E}^1, \dots, \mathcal{E}^m)$ . Since the feasible set is convex, there exists a unique maximizer in total efforts per project  $(\mathcal{E}^{1*}, \dots, \mathcal{E}^{m*})$ . Now, take any optimal project assignment  $\{\{\hat{e}_i^p\}_{p=1}^m\}_{i=1}^n$  and consider the average assignment:

$$\bar{e}_i^p := \bar{e}^{p*} = \frac{1}{n} \sum_{j=1}^n \hat{e}_j^p$$

The total efforts at each project remain unchanged, and so  $\{\{\bar{e}_i^p\}_{p=1}^m\}_{i=1}^n$  also achieves the principal's maximum. To show it is feasible, note that since  $\{\{\hat{e}_i^p\}_{p=1}^m\}_{i=1}^n$  is feasible, then:

$$\alpha + g \sum_{j \neq i} \hat{e}_j^p \geq c \hat{e}_i^p + r \sum_{q \neq p} \hat{e}_i^q \implies n\alpha + g \sum_{j \neq i} \sum_{\ell=1}^n \hat{e}_\ell^p \geq c \sum_{j=1}^n \hat{e}_j^p + r \sum_{q \neq p} \sum_{j=1}^n \hat{e}_i^q$$



Now we need to show the incentive constraints for  $\{\{\bar{e}_i^p\}_{p=1}^m\}_{i=1}^n$  also holds. Take any incentive constraint.

$$\begin{aligned}
\alpha + g \sum_{j \neq i} \bar{e}_j^p - c \bar{e}_i^p - r \sum_{q \neq p} \bar{e}_i^q &= \alpha + g \sum_{j \neq i} \left( \frac{1}{n} \sum_{\ell=1}^n \hat{e}_\ell^p \right) - c \left( \frac{1}{n} \sum_{j=1}^n \hat{e}_j^p \right) - r \sum_{q \neq p} \left( \frac{1}{n} \sum_{j=1}^n \hat{e}_j^q \right) \\
&= \alpha + g \left( \frac{n-1}{n} \sum_{\ell=1}^n \hat{e}_\ell^p \right) - c \left( \frac{1}{n} \sum_{j=1}^n \hat{e}_j^p \right) - r \sum_{q \neq p} \left( \frac{1}{n} \sum_{j=1}^n \hat{e}_j^q \right) \\
&= n\alpha + g \left( (n-1) \sum_{\ell=1}^n \hat{e}_\ell^p \right) - c \left( \sum_{j=1}^n \hat{e}_j^p \right) - r \sum_{q \neq p} \left( \sum_{j=1}^n \hat{e}_j^q \right) \\
&= n\alpha + g \sum_{j \neq i} \sum_{\ell=1}^n \hat{e}_\ell^p - c \sum_{j=1}^n \hat{e}_j^p - r \sum_{q \neq p} \sum_{j=1}^n \hat{e}_j^q \\
&\geq 0
\end{aligned}$$

(1) In the worker-symmetric optimal project assignment, every worker plays the same optimal effort at every project. Because the Cobb-Douglas objective satisfies Inada conditions, every worker must play positive effort at every project.

(2) The worker-symmetric optimal project assignment solves the symmetric problem:

$$\begin{aligned}
&\max_{\{e^p\}_{p=1}^m} \sum_{p=1}^m \beta^{(p)} \ln(e^p) \\
&\text{s.t. } \alpha + g(n-1)e^p \geq ce^p + r \sum_{q \neq p} e^q
\end{aligned}$$

A unique maximizer  $\{e^{p*}\}_{p=1}^m$  exists, and the KKT conditions are both necessary and sufficient. Consider project  $s$  and rewrite the incentive constraint in terms of the total effort in the entire system:

$$\alpha + g(n-1)e^s \geq ce^s + r \sum_{p \neq s} e^p \implies \alpha + (g(n-1) - c + r)e^s \geq r \sum_{p=1}^m e^p$$

For any projects  $s$  at which the incentive constraint binds, we have:

$$\alpha + (g(n-1) - c + r)e^{s*} = r \sum_{p=1}^m e^{p*} \implies e^{s*} = \frac{\alpha - r \sum_{p=1}^m e^{p*}}{c - r - g(n-1)}$$

For project  $t$  with a slack incentive constraint, we have

$$\alpha + (g(n-1) - c + r)e^{t*} > \sum_{q=1}^m e^{q*} \implies e^{t*} < \frac{\alpha - r \sum_{q=1}^m e^{q*}}{c - r - g(n-1)}$$

This means  $e^{t*} < e^{s*}$  for any project  $t$  and  $s$  with slack and binding incentive constraints, respectively.

All projects with binding incentive constraints have the same optimal effort, which depends on the total effort in the entire system. The first-order condition for any project  $s$  with a binding incentive constraint is

$$\frac{\beta^{(s)}}{e^{s*}} = \lambda^{s*}(c - g(n-1) - r) + r \sum_{p=1}^m \lambda^{p*} \geq r \sum_{p=1}^m \lambda^{p*}$$

For project  $t$  with a slack incentive constraint ( $\lambda^{t*} = 0$ ), its first-order condition is:

$$\frac{\beta^{(t)}}{e^{t*}} + \lambda^{t*}(g(n-1) - c + r) - r \sum_{q=1}^m \lambda^{q*} = 0 \implies \frac{\beta^{(t)}}{e^{t*}} = r \sum_{p=1}^m \lambda^{p*}$$

Linking both inequalities together yields:

$$\beta^{(s)} \geq e^{s*} \left( r \sum_{p=1}^m \lambda^{p*} \right) > e^{t*} \left( r \sum_{p=1}^m \lambda^{p*} \right) = \beta^{(t)}$$

Every project with a binding incentive constraint has a strictly larger  $\beta$ -weight than every project with a slack incentive constraint. Since projects are ordered  $\beta^{(1)} > \dots > \beta^{(m)}$ , that means there exists a project index  $\zeta^*$  that for all projects  $s \leq \zeta^*$ ,  $\beta^{(s)} \geq \beta^{(\zeta^*)}$  with a binding incentive constraint and for all projects  $t > \zeta^*$ ,  $\beta^{(t)} < \beta^{(\zeta^*)}$  with a slack incentive constraint. The formula for  $\zeta^*$  is derived in (4).

(3) I compute the optimal total effort in the entire system, which is a sufficient statistic for efforts at projects with a binding incentive constraint. All projects with binding incentive constraints have the same optimal effort. The following substitutes in the first-order condition for projects with slack incentive constraints.

$$\sum_{p=1}^m e^{p*} = \sum_{p=1}^{\zeta^*} e^{p*} + \sum_{t=\zeta^*+1}^m e^{t*} = \zeta^* e^{s*} + \left( \frac{1}{r \sum_{p=1}^m \lambda^{p*}} \right) \sum_{t=\zeta^*+1}^m \beta^{(t)}$$

From the binding incentive constraint on project  $s$ , we obtain:

$$\alpha + (g(n-1) - c + r)e^{s*} = r \sum_{p=1}^m e^{p*} \implies \alpha = (c - r - g(n-1) + r\zeta^*)e^{s*} + \left( \frac{1}{\sum_{p=1}^m \lambda^{p*}} \right) \sum_{t=q^*+1}^m \beta^{(t)}$$

Next, we determine a relationship between the multipliers  $\sum_{p=1}^m \lambda^{p*}$  and the relative project weights  $\beta$ . Summing the first-order conditions for all projects  $s$  with binding incentive constraints gives:

$$\frac{\beta^{(s)}}{e^{s*}} = (c - r - g(n-1))\lambda^{s*} + r \sum_{p=1}^m \lambda^{p*} \implies \sum_{p=1}^{\zeta^*} \frac{\beta^{(p)}}{e^{p*}} = (c - r - g(n-1)) \sum_{p=1}^{\zeta^*} \lambda^{p*} + \zeta^* r \sum_{p=1}^m \lambda^{p*}$$

Note that  $\sum_{p=1}^{\zeta^*} \lambda^{p*} = \sum_{p=1}^m \lambda^{p*}$  because  $\lambda^{t*} = 0$  for all  $t \geq \zeta^*$ . Together with the fact that all efforts at projects with binding incentive constraints are identical, we can write:

$$\frac{1}{e^{s*}} \sum_{p=1}^{\zeta^*} \beta^{(p)} = (c - r - g(n-1)) + \zeta^* r \sum_{p=1}^m \lambda^{p*} \implies e^{s*}(c - r - g(n-1)) + \zeta^* r = \frac{\sum_{p=1}^{\zeta^*} \beta^{(p)}}{\sum_{p=1}^m \lambda^{p*}}$$

Finally, substitute this into the equation for  $\alpha$ :

$$\alpha = \frac{\sum_{p=1}^{\zeta^*} \beta^{(p)}}{\sum_{p=1}^m \lambda^{p*}} + \left( \frac{1}{\sum_{p=1}^m \lambda^{p*}} \right) \sum_{t=\zeta^*+1}^m \beta^{(t)} \implies \sum_{p=1}^m \lambda^{p*} = \frac{\sum_{p=1}^m \beta^{(p)}}{\alpha}$$

Now we can directly compute  $e^{s*}$  and  $e^{t*}$ :

$$e^{s*} = \frac{\alpha - \left( \frac{1}{\sum_{p=1}^m \lambda^{p*}} \right) \sum_{t=\zeta^*+1}^m \beta^{(t)}}{c - r - g(n-1) + r\zeta^*} \implies e^{s*} = \frac{\alpha}{c - r - g(n-1) + \zeta^* r} \left( \frac{\sum_{p=1}^{\zeta^*} \beta^{(p)}}{\sum_{p=1}^m \beta^{(p)}} \right)$$

$$e^{t*} = \frac{\beta^{(t)}}{r \sum_{p=1}^m \lambda^{p*}} = \frac{\alpha}{r} \left( \frac{\beta^{(s)}}{\sum_{p=1}^m \beta^{(p)}} \right)$$

**(4)** For project  $s$  with a binding incentive constraint, the first-order condition in-

equality with substituting in the closed form for  $e^{s*}$  gives:

$$\frac{\beta^{(s)}}{e^{s*}} \geq r \sum_{p=1}^m \lambda^{p*} \implies \beta^{(s)} \left( \frac{c - r - g(n-1) + \zeta^* r}{\alpha} \right) \left( \frac{\sum_{p=1}^m \beta^{(p)}}{\sum_{p=1}^{\zeta^*} \beta^{(p)}} \right) \geq r \left( \frac{\sum_{p=1}^m \beta^{(p)}}{\alpha} \right)$$

The right-hand inequality simplifies to

$$\frac{\beta^{(s)}}{r} \geq \frac{\sum_{p=1}^{\zeta^*} \beta^{(p)}}{c - r - g(n-1) + \zeta^* r}$$

Since this inequality holds for all  $s \leq \zeta^*$ , it must be that

$$\frac{\beta^{\zeta^*}}{r} \geq \frac{\sum_{p=1}^{\zeta^*} \beta^{(p)}}{c - r - g(n-1) + \zeta^* r}$$

Note that project  $\zeta^* + 1$  must have a slack incentive constraint. We can substitute the formulas for  $e^{q*}$  and  $e^{q*+1}$  into the inequality  $e^{\zeta^*+1} < e^{q*}$ :

$$\frac{\alpha}{r} \left( \frac{\beta^{(\zeta^*+1)}}{\sum_{p=1}^m \beta^{(p)}} \right) < \frac{\alpha}{c - r - g(n-1) + \zeta^* r} \left( \frac{\sum_{p=1}^{\zeta^*} \beta^{(p)}}{\sum_{p=1}^m \beta^{(p)}} \right) \implies \frac{\beta^{\zeta^*+1}}{r} < \frac{\sum_{p=1}^{\zeta^*} \beta^{(p)}}{c - r - g(n-1) + \zeta^* r}$$

Thus  $\zeta^*$  is a natural number that satisfies:

$$\frac{\beta^{(\zeta^*+1)}}{r} < \frac{\sum_{p=1}^{\zeta^*} \beta^{(p)}}{c - r - g(n-1) + \zeta^* r} \leq \frac{\beta^{(\zeta^*)}}{r}$$

If there was another natural number  $\hat{q} \neq \zeta^*$  that satisfies the above thresholds, then  $\hat{q}$  induce  $\hat{e}^s \neq e^{s*}$  and  $\hat{e}^t \neq e^{t*}$  for all projects with binding and slack incentive constraints, respectively. However, this contradicts the uniqueness of the optimal  $e^*$ . From the above, we can immediately see that  $\zeta^*$  is the maximum project index that satisfies

$$\frac{\sum_{p=1}^{\zeta} \beta^{(p)}}{c - r - g(n-1) + \zeta r} \leq \frac{\beta^{(\zeta)}}{r} \iff \frac{\sum_{p=1}^{\zeta} \beta^{(p)}}{\beta^{(\zeta)}} - \zeta \leq \frac{c - r - g(n-1)}{r}.$$

I show that  $\zeta^*$  is non-increasing in  $r$  using the established bounds about  $\zeta^*$ . From

the right inequality, we have the equivalent statement

$$\frac{\sum_{p=1}^{\zeta^*} \beta^{(p)}}{c - r - g(n-1) + \zeta^* r} \leq \frac{\beta^{(\zeta^*)}}{r} \iff r \left( \sum_{p=1}^{\zeta^*} \beta^{(p)} - (\zeta^* - 1) \beta^{(\zeta^*)} \right) \leq \beta^{(\zeta^*)} (c - g(n-1))$$

Because we ordered  $\beta^{(1)} > \dots > \beta^{(m)}$ ,

$$\sum_{p=1}^{\zeta^*} \beta^{(p)} - \zeta^* \beta^{(\zeta^*)} > 0 \implies \sum_{p=1}^{\zeta^*} \beta^{(p)} - (\zeta^* - 1) \beta^{(\zeta^*)} > 0$$

We derive the following upper bound on  $r$  in terms of  $\zeta^*$ :

$$r \leq \frac{\beta^{(\zeta^*)} (c - g(n-1))}{\sum_{p=1}^{\zeta^*} \beta^{(p)} - (\zeta^* - 1) \beta^{(\zeta^*)}} := U_{\zeta^*}$$

From the left inequality that  $\zeta^*$  satisfies in (4a), we have the equivalent statement:

$$\frac{\beta^{(\zeta^*+1)}}{r} < \frac{\sum_{p=1}^{\zeta^*} \beta^{(p)}}{c - r - g(n-1) + \zeta^* r} \iff r \left( \sum_{p=1}^{\zeta^*} \beta^{(p)} - (\zeta^* - 1) \beta^{(\zeta^*+1)} \right) > \beta^{(\zeta^*+1)} (c - g(n-1))$$

This establishes a lower bound on  $r$  in terms of  $\zeta^*$ :

$$r > \frac{\beta^{(\zeta^*+1)} (c - g(n-1))}{\sum_{p=1}^{\zeta^*} \beta^{(p)} - (\zeta^* - 1) \beta^{(\zeta^*+1)}} := L_{\zeta^*}$$

Note that  $U_{q+1} = L_q$  and  $U_q$  is strictly decreasing in  $q$ . I show this by computing the discrete difference

$$U_{q+1} - U_q = \frac{(\beta^{(q+1)} - \beta^{(q)}) (c - g(n-1)) \sum_{p=1}^q \beta^{(p)}}{(\sum_{p=1}^q \beta^{(p)} - (q-1) \beta^{(q)}) (\sum_{p=1}^q \beta^{(p)} - (q-1) \beta^{(q+1)})} < 0$$

Thus we have partitioned the  $r$ -space into  $m$  disjoint intervals:

$$[L_m, U_m) \cup [L_{m-1}, U_{m-1}) \cup \dots \cup [L_1, U_1)$$

where  $L_m = 0$  and  $U_1 = c - g(n-1)$ . For any  $r$  that satisfies the strict diagonal dominance condition i.e.  $r \in [0, c - g(n-1))$ , there exists a unique  $\zeta^*$  such that

$L_{\zeta^*} \leq r < U_{\zeta^*}$ . Thus  $\zeta^*$  is a non-increasing step function of  $r$  given by:

$$\zeta^*(r) = \max\{q \mid r < U_q\}$$

The bounds  $U_q$  are increasing in  $c$  and decreasing in  $g$  and  $n$ . Therefore,  $\zeta^*$  is a non-decreasing step function in  $c$  and a non-increasing step function in  $g$  and  $n$ . Because  $U_q$  is independent of  $\alpha$ , then  $\zeta^*$  is also independent of  $\alpha$ .  $\square$

## A.7 Proof of Theorem 3

I begin by proving the ordering of the project marginal values.

**Lemma 2.** *In any optimum,  $\frac{\beta^1}{\mathcal{E}^{1*}} \geq \dots \geq \frac{\beta^m}{\mathcal{E}^{m*}}$ .*

*Proof.* I perform a column swap on the dual problem to reach a contradiction. Writing out the Lagrangian

$$\mathcal{L} = \sum_{p=1}^m \beta^{(p)} \ln(\mathcal{E}^p) + \sum_{p=1}^m \sum_{i=1}^n \lambda_i^p (\alpha + g \sum_{j \neq i} e_j^p - c e_i^p - r_i \sum_{q \neq p} e_i^q)$$

Define the multiplier sums  $\Lambda_i := \sum_{p=1}^m \lambda_i^p$  and  $\Lambda^p := \sum_{i=1}^n \lambda_i^p$ . I rewrite the Lagrangian as

$$\begin{aligned} \mathcal{L} &= \sum_{p=1}^m \beta^{(p)} \ln(\mathcal{E}^p) + \sum_{p=1}^m \sum_{i=1}^n e_i^p ((r_i - c - g) \lambda_i^p + g \Lambda^p - r_i \Lambda_i) + \sum_{p=1}^m \sum_{i=1}^n \alpha \lambda_i^p \\ &= \sum_{p=1}^m \beta^{(p)} \ln(\mathcal{E}^p) + \sum_{p=1}^m \mathcal{E}^p g \Lambda^p + \sum_{p=1}^m \sum_{i=1}^n e_i^p ((r_i - c - g) \lambda_i^p - r_i \Lambda_i) + \sum_{p=1}^m \sum_{i=1}^n \alpha \lambda_i^p \end{aligned}$$

To form the dual, I maximize the Lagrangian over all  $\mathbf{e}$  given a fixed  $\boldsymbol{\lambda} \geq \mathbf{0}$ . The maximization problem is separable across projects; the following is it for project  $p$ :

$$\max_{\{e_i^p\}_{i=1}^n} \beta^{(p)} \ln(\mathcal{E}^p) + \mathcal{E}^p g \Lambda^p + \sum_{i=1}^n e_i^p ((r_i - c - g) \lambda_i^p - r_i \Lambda_i)$$

Observe that  $((r_i - c - g) \lambda_i^p - r_i \Lambda_i) \leq 0$  for all  $i$ . Then, fixing the total effort  $\mathcal{E}^p$ , the objective is maximized by setting  $e_i^p = \mathcal{E}^p$  for worker  $i$  with the maximum

$((r_i - c - g)\lambda_i^p - r_i\Lambda_i)$  term. Therefore, we can rewrite the maximization problem

$$\max_{\mathcal{E}^p} \beta^{(p)} \ln(\mathcal{E}^p) + \mathcal{E}^p g \Lambda^p + \mathcal{E}^p \max_i \{(r_i - c - g)\lambda_i^p - r_i\Lambda_i\}$$

The unique maximizer is

$$\mathcal{E}^{p*} = \frac{\beta^{(p)}}{-\max_i \{(r_i - c - g)\lambda_i^p - r_i\Lambda_i\} - g\Lambda^p}$$

With some cancellations, we can form the dual function

$$D(\lambda) = \sum_{p=1}^m \left( \beta^{(p)} \ln(\mathcal{E}^{p*}) - \beta^{(p)} \right) + \sum_{p=1}^m \sum_{i=1}^n \alpha \lambda_i^p$$

Slater's condition for the primal holds, implying strong duality. Now assume by contradiction that  $\lambda^*$  minimizes the dual and that there exists  $p < q$  with  $\frac{\beta^p}{\mathcal{E}^{p*}} < \frac{\beta^q}{\mathcal{E}^{q*}}$ . Construct a swap  $\tilde{\lambda}$  by setting:

$$\tilde{\lambda}_i^p = \lambda_i^{q*} ; \quad \tilde{\lambda}_i^q = \lambda_i^{p*} ; \quad \tilde{\lambda}_i^s = \lambda_i^{s*} \quad \forall s \neq p, q$$

Observe that  $\tilde{\Lambda}_i = \Lambda_i^*$  for all workers  $i$ , however the project multiplier sums for  $p$  and  $q$  are swapped i.e.  $\tilde{\Lambda}^p = \Lambda^{q*}$  and  $\tilde{\Lambda}^q = \Lambda^{p*}$ . This means that for project  $p$ ,

$$\max_i \{(r_i - c - g)\tilde{\lambda}_i^p - r_i\tilde{\Lambda}_i\} = \max_i \{(r_i - c - g)\lambda_i^{q*} - r_i\Lambda_i^*\}$$

The same swap occurs for project  $q$ . Now compute:

$$\begin{aligned} D(\tilde{\lambda}) - D(\lambda^*) &= -\beta^{(p)} \ln(-\max_i \{(r_i - c - g)\lambda_i^{q*} - r_i\Lambda_i^*\} - g\Lambda^{q*}) \\ &\quad - \beta^{(q)} \ln(-\max_i \{(r_i - c - g)\lambda_i^{p*} - r_i\Lambda_i^*\} - g\Lambda^{p*}) + \alpha \left( \sum_{i=1}^n \lambda_i^{p*} + \sum_{i=1}^n \lambda_i^{q*} \right) \\ &\quad + \beta^{(p)} \ln(-\max_i \{(r_i - c - g)\lambda_i^{p*} - r_i\Lambda_i^*\} - g\Lambda^{p*}) \\ &\quad + \beta^{(q)} \ln(-\max_i \{(r_i - c - g)\lambda_i^{q*} - r_i\Lambda_i^*\} - g\Lambda^{q*}) - \alpha \left( \sum_{i=1}^n \lambda_i^{q*} + \sum_{i=1}^n \lambda_i^{p*} \right) \end{aligned}$$

Combining like terms yields

$$\beta^p \ln \left( \frac{-\max_i \{(r_i - c - g)\lambda_i^{p*} - r_i \Lambda_i^*\} - g\Lambda^{p*}}{-\max_i \{(r_i - c - g)\lambda_i^{q*} - r_i \Lambda_i^*\} - g\Lambda^{q*}} \right) + \beta^q \ln \left( \frac{-\max_i \{(r_i - c - g)\lambda_i^{q*} - r_i \Lambda_i^*\} - g\Lambda^{q*}}{-\max_i \{(r_i - c - g)\lambda_i^{p*} - r_i \Lambda_i^*\} - g\Lambda^{p*}} \right)$$

Recall the unique maximizer  $\mathcal{E}^{p*}$  satisfies

$$\frac{\beta^{(p)}}{\mathcal{E}^{p*}} = -\max_i \{(r_i - c - g)\lambda_i^{p*} - r_i \Lambda_i^*\} - g\Lambda^{p*}$$

Substituting back in, we obtain

$$\beta^{(p)} \ln \left( \frac{\beta^{(p)} / \mathcal{E}^{p*}}{\beta^{(q)} / \mathcal{E}^{q*}} \right) + \beta^{(q)} \ln \left( \frac{\beta^{(q)} / \mathcal{E}^{q*}}{\beta^{(p)} / \mathcal{E}^{p*}} \right) = (\beta^{(p)} - \beta^{(q)}) \ln \left( \frac{\beta^{(p)} / \mathcal{E}^{p*}}{\beta^{(q)} / \mathcal{E}^{q*}} \right) < 0$$

The inequality holds because  $p < q \implies \beta^{(p)} - \beta^{(q)} > 0$  and our original assumption that  $\frac{\beta^{(p)}}{\mathcal{E}^{p*}} < \frac{\beta^{(q)}}{\mathcal{E}^{q*}}$ . But this implies  $D(\tilde{\lambda}) - D(\lambda^*) < 0$ , which contradicts minimality of  $\lambda^*$ . Thus we must have  $\frac{\beta^{(p)}}{\mathcal{E}^{p*}} \geq \frac{\beta^{(q)}}{\mathcal{E}^{q*}}$ .  $\square$

Since  $\frac{\partial V}{\partial \mathcal{E}^p} = \frac{\beta^{(p)}}{\mathcal{E}^p}$ , we can apply [Theorem 1](#) to conclude that  $A^p$  and  $\mathcal{F}^p$  both obey the same project ordering given by the project weights  $\beta^{(p)}$ . Thus, there exists a worker-specific project index  $\gamma_i^*$  such that  $e_i^{p*} > 0$  if and only if  $p \leq \gamma_i^*$ . To relate  $\gamma_i^*$  to  $r_i$ , I explicitly solve for  $\lambda_i^p$  and  $\mu_i^p$  by the following case analysis. Note the linear complementarity problem that the multipliers solve is given by

$$\mu^p = \mathbf{M}\lambda^p + \mathbf{R}\Lambda - \left( \frac{\beta^{(p)}}{\mathcal{E}^p} \right) \mathbf{1}_n; \lambda^p, \mu^p \geq \mathbf{0}; (\lambda^p)^\top \mu^p = \mathbf{0}$$

For notation, let  $\Lambda^p := \sum_{i=1}^n \lambda_i^p$ .

Case #1.1:  $e_i^p > 0$ .

Then  $\mu_i^p = 0$ . From the linear complementarity problem, we have

$$0 = \lambda_i^p(c + g - r_i) + r_i \Lambda_i - \frac{\beta^{(p)}}{\mathcal{E}^p} - g\Lambda^p = 0 \implies \lambda_i^p = \frac{\frac{\beta^{(p)}}{\mathcal{E}^p} + g\Lambda^p - r_i \Lambda_i}{c + g - r_i}$$

Since  $\lambda_i^p \geq 0$ , it must be that  $\frac{\beta^{(p)}}{\mathcal{E}^p} + g\Lambda^p \geq r_i \Lambda_i$ .



Case #1.2:  $e_i^p = 0$ .

Then  $\lambda_i^p = 0$  by the generic strict slackness assumption. We have

$$\mu_i^p = r_i \Lambda_i - \frac{\beta^{(p)}}{\varepsilon^p} - g \Lambda^p$$

Putting both cases together, we have the following expressions for  $\lambda_i^p$  and  $\mu_i^p$ :

$$\lambda_i^p = \frac{\max(\frac{\beta^{(p)}}{\varepsilon^p} + g \Lambda^p - r_i \Lambda_i, 0)}{c + g - r_i} \quad \mu_i^p = \max(r_i \Lambda_i - \frac{\beta^{(p)}}{\varepsilon^p} - g \Lambda^p, 0)$$

If  $\frac{\beta^{(p)}}{\varepsilon^p} + g \Lambda^p > r_i \Lambda_i$ , then  $\lambda_i^p > 0$ . By [Assumption 2](#),  $\lambda_i^p > 0$  implies  $e_i^p > 0$ . If  $\frac{\beta^{(p)}}{\varepsilon^p} + g \Lambda^p < r_i \Lambda_i$ , then  $\mu_i^p > 0$ , forcing  $e_i^p = 0$ . Finally, if  $\frac{\beta^{(p)}}{\varepsilon^p} + g \Lambda^p = r_i \Lambda_i$ , then both  $\lambda_i^p = \mu_i^p = 0$ . This allows us to write  $\gamma_i^* = \max\{p \mid \frac{\beta^{(p)}}{\varepsilon^p} + g \Lambda^p \geq r_i \Lambda_i\}$  and also shows that  $\frac{\beta^{(p)}}{\varepsilon^p} + g \Lambda^p$  is non-increasing in  $p$ .

Now I show that  $r_i > r_j$  implies  $\sigma_i^* \leq \sigma_j^*$ . This is equivalent to showing that  $r_i \Lambda_i \geq r_j \Lambda_j$ . Assume by contradiction that  $r_i \Lambda_i < r_j \Lambda_j$ , and so  $r_j \Lambda_j > 0$  necessarily. There must exist at least one project with  $\frac{\beta^{(p)}}{\varepsilon^p} + g \Lambda^p > r_j \Lambda_j$ ; otherwise if  $\frac{\beta^{(p)}}{\varepsilon^p} + g \Lambda^p \leq r_j \Lambda_j$  for all  $p$ , then  $\mu_j^p > 0$  for all  $p$ , forcing  $e_j^p = 0$ . By the generic strict slackness assumption,  $\lambda_j^p = 0$  for all  $p$ , implying  $\Lambda_j = 0$ . But this contradicts  $r_j \Lambda_j > 0$ . Therefore, there exists some project  $p$  such that  $\frac{\beta^{(p)}}{\varepsilon^p} + g \Lambda^p > r_j \Lambda_j > r_i \Lambda_i$ , implying

$$\lambda_i^p = \frac{\frac{\beta^{(p)}}{\varepsilon^p} + g \Lambda^p - r_i \Lambda_i}{c + g - r_i} > 0; \quad \lambda_j^p = \frac{\frac{\beta^{(p)}}{\varepsilon^p} + g \Lambda^p - r_j \Lambda_j}{c + g - r_j} > 0$$

Since  $\frac{\beta^{(p)}}{\varepsilon^p} + g \Lambda^p - r_i \Lambda_i > \frac{\beta^{(p)}}{\varepsilon^p} + g \Lambda^p - r_j \Lambda_j$  and  $c + g - r_i < c + g - r_j$ , then  $\lambda_i^p > \lambda_j^p$  for all  $p$  with the property  $\frac{\beta^{(p)}}{\varepsilon^p} + g \Lambda^p > r_j \Lambda_j$ .

Now consider projects  $q$  with  $\frac{\beta^{(q)}}{\varepsilon^q} + g \Lambda^q \in (r_i \Lambda_i, r_j \Lambda_j]$ , which have the property

$\lambda_i^q > 0 = \lambda_j^q$ . Summing the  $\lambda$  multipliers across all projects.

$$\begin{aligned}\Lambda_i &= \sum_{p \mid \frac{\beta^{(p)}}{\mathcal{E}^p} + g\Lambda^p > r_j\Lambda_j} \lambda_i^p + \underbrace{\sum_{p \mid r_i\Lambda_i < \frac{\beta^{(p)}}{\mathcal{E}^p} + g\Lambda^p \leq r_j\Lambda_j} \lambda_i^p}_{=0} + \underbrace{\sum_{p \mid \frac{\beta^{(p)}}{\mathcal{E}^p} + g\Lambda^p \leq r_i\Lambda_i} \lambda_i^p}_{=0} \\ \Lambda_j &= \sum_{p \mid \frac{\beta^{(p)}}{\mathcal{E}^p} + g\Lambda^p > r_j\Lambda_j} \lambda_j^p + \underbrace{\sum_{p \mid r_i\Lambda_i < \frac{\beta^{(p)}}{\mathcal{E}^p} + g\Lambda^p \leq r_j\Lambda_j} \lambda_j^p}_{=0} + \underbrace{\sum_{p \mid \frac{\beta^{(p)}}{\mathcal{E}^p} + g\Lambda^p \leq r_i\Lambda_i} \lambda_j^p}_{=0}\end{aligned}$$

We obtain  $\Lambda_i > \Lambda_j \implies r_i\Lambda_i > r_j\Lambda_j$ , which is a direct contradiction.

Because  $\frac{\beta^{(p)}}{\mathcal{E}^p} + g\Lambda^p$  is non-increasing in  $p$ , the active set for worker  $i$  is  $\{p \mid \frac{\beta^{(p)}}{\mathcal{E}^p} + g\Lambda^p \geq r_i\Lambda_i\}$ . Thereby  $r_i > r_j \implies r_i\Lambda_i \geq r_j\Lambda_j \implies \{p \mid \frac{\beta^{(p)}}{\mathcal{E}^p} + g\Lambda^p \geq r_i\Lambda_i\} \subseteq \{p \mid \frac{\beta^{(p)}}{\mathcal{E}^p} + g\Lambda^p \geq r_j\Lambda_j\} \implies \sigma_j^* \leq \sigma_i^*$ . Similarly, we can conclude that  $\gamma_j^* \leq \gamma_i^*$ .  $\square$

## A.8 Proof of Theorem 4

The proof that the marginal project valuations for the principal satisfy  $\frac{\beta^{(1)}}{\mathcal{E}^{1*}} \geq \dots \geq \frac{\beta^{(m)}}{\mathcal{E}^{m*}}$  is almost identical to that in Lemma 2 and thus I do not replicate it here. By the argument in Appendix B, we operate under the generic strict slackness assumption that  $e_i^p = 0$  implies  $\lambda_i^p = 0$  in the optimum. Like before, we have the linear complementarity problem that the multipliers solve, given by

$$\boldsymbol{\mu}^p = \mathbf{M}\boldsymbol{\lambda}^p + \mathbf{R}\boldsymbol{\Lambda} - \left(\frac{\beta^{(p)}}{\mathcal{E}^p}\right) \mathbf{1}_n; \boldsymbol{\lambda}^p, \boldsymbol{\mu}^p \geq \mathbf{0}; (\boldsymbol{\lambda}^p)^\top \boldsymbol{\mu}^p = 0$$

To understand when workers are unassigned from projects due to a decreasing marginal value of projects  $\left(\frac{\beta^{(p)}}{\mathcal{E}^p}\right)$ , which is a discrete parameter, I turn to its continuous counterpart.

$$\boldsymbol{\mu}^p = \mathbf{M}\boldsymbol{\lambda}^p + \mathbf{R}\boldsymbol{\Lambda} - \rho \mathbf{1}_n; \boldsymbol{\lambda}^p, \boldsymbol{\mu}^p \geq \mathbf{0}; (\boldsymbol{\lambda}^p)^\top \boldsymbol{\mu}^p = 0$$

$\left(\frac{\beta^{(p)}}{\mathcal{E}^p}\right)$  is replaced by  $\rho \in \mathbb{R}$ . I employ a pivot-type method. Ignoring non-negativity of  $\lambda^p$ , I solve the system assuming that  $\mu^p = \mathbf{0}$ , i.e., all workers are active on project  $p$ . We obtain:

$$\mathbf{M}\lambda^p + \mathbf{R}\Lambda - \sigma\mathbf{1}_n = \mathbf{0} \implies \lambda^p = \mathbf{M}^{-1}(\rho\mathbf{1}_n - \mathbf{R}\Lambda)$$

If  $\rho$  is sufficiently large, then  $\lambda^p > \mathbf{0}$  and all workers are assigned positive effort on project  $p$ . As  $\rho$  decreases,  $\lambda^p$  likewise decreases. The first index  $i$  of  $\lambda^p$  to reach zero is when

$$\rho = \left( \frac{r(\mathbf{M}^{-1}\Lambda)_i}{(\mathbf{M}^{-1}\mathbf{1})_i} \right)$$

and it becomes negative if

$$\rho < \left( \frac{r(\mathbf{M}^{-1}\Lambda)_i}{(\mathbf{M}^{-1}\mathbf{1})_i} \right)$$

Thus the index  $i$  corresponding to  $\arg \max_i \left( \frac{r(\mathbf{M}^{-1}\Lambda)_i}{(\mathbf{M}^{-1}\mathbf{1})_i} \right)$  is the first worker to be assigned  $\mu_i^p > 0$ , else  $\lambda_i^p < 0$  which contradicts the latter's nonnegativity. This implies worker  $i$  must be unassigned from all projects  $p$  with

$$\frac{\beta^{(p)}}{\mathcal{E}^p} < \left( \frac{r(\mathbf{M}^{-1}\Lambda)_i}{(\mathbf{M}^{-1}\mathbf{1})_i} \right)$$

On those projects, we can employ the same strategy, but now restrict attention to the first index  $j$  of  $\lambda^{p, N \setminus \{i\}}$  to reach zero, which is when

$$\rho = \left( \frac{r((\mathbf{M}^{N \setminus \{i\}})^{-1}\Lambda^{N \setminus \{i\}})_i}{((\mathbf{M}^{N \setminus \{i\}})^{-1}\mathbf{1}^{N \setminus \{i\}})_j} \right)$$

This is precisely because worker  $i$  is no longer an active participant on those projects. The same logic extends recursively to all remaining workers.  $\square$

## A.9 Proof of Theorem 5

Under full utilization and activity, the optimal project assignment has, for each worker  $i$  and project  $p$ ,  $e_i^{p*} > 0$ , and the incentive constraint binds

$$\alpha_i^p + \sum_{j \neq i} g_{ji} e_j^{p*} = c_i e_i^{p*} + r_i \sum_{q \neq p} e_i^{q*}$$

Therefore, we have

$$\alpha^p = \mathbf{M} \mathbf{e}^p + \mathbf{R} \mathbf{E} \mathbf{1}_m$$

Now sum over all projects to obtain

$$\sum_{p=1}^m \alpha^p = \sum_{p=1}^m (\mathbf{M} \mathbf{e}^p + \mathbf{R} \mathbf{E} \mathbf{1}_m) \implies \sum_{p=1}^m \alpha^p = \mathbf{M} \mathbf{E} \mathbf{1}_m + m \mathbf{R} \mathbf{E} \mathbf{1}_m$$

Therefore the total effort vector satisfies  $\mathbf{E} \mathbf{1}_m = \mathbf{P}^{-1} \sum_{p=1}^m \alpha^p$ . Substituting into the incentive constraint for project  $p$  gives

$$\alpha^p = \mathbf{M} \mathbf{e}^p + \mathbf{R} \mathbf{P}^{-1} \sum_{p=1}^m \alpha^p \mathbf{1}_m \implies \mathbf{e}^p = \mathbf{M}^{-1} \alpha^p - \mathbf{M}^{-1} \mathbf{R} \mathbf{P}^{-1} \sum_{p=1}^m \alpha^p$$

Now we can differentiate

$$\frac{\partial \mathbf{e}^p}{\partial \alpha_j^q} = \mathbf{M}^{-1} \epsilon_j \delta_{pq} - \mathbf{M}^{-1} \mathbf{R} \mathbf{P}^{-1} \epsilon_j$$

Reading off the  $i$ -th entry gives the formula in the theorem. All of the other equations are derived similarly.

To show the sign of  $\frac{\partial e_i^{p*}}{\partial \alpha_j^q}$ , note that its matrix form is given by:

$$\frac{\partial \mathbf{e}^p}{\partial \alpha^q} = \mathbf{M}^{-1} \delta_{pq} - \mathbf{M}^{-1} \mathbf{R} \mathbf{P}^{-1} = \mathbf{M}^{-1} \delta_{pq} - \frac{1}{m} (\mathbf{M}^{-1} - \mathbf{P}^{-1})$$

The second equality hold because  $\mathbf{R} = \frac{1}{m} (\mathbf{P} - \mathbf{M})$ . Therefore:

$$\frac{\partial \mathbf{e}^p}{\partial \alpha^q} = \begin{cases} \frac{m-1}{m} \mathbf{M}^{-1} + \frac{1}{m} \mathbf{P}^{-1} & p = q \\ -\frac{1}{m} (\mathbf{M}^{-1} - \mathbf{P}^{-1}) & p \neq q \end{cases}$$

Therefore  $\frac{\partial e_i^{p*}}{\partial \alpha_j^p} > 0$  for all  $j \neq i$  and  $\frac{\partial e_i^{p*}}{\partial \alpha_j^q} < 0$  for all  $q \neq p$ .  $\square$

## A.10 Proof of Theorem 6

As an abuse of notation,  $\mathbf{e}^p$  and  $\boldsymbol{\alpha}^p$  both refer to the efforts and preferences of workers restricted to those in  $A^p$ . Define  $\mathbf{E} = [\mathbf{e}^1 \cdots \mathbf{e}^n]$ . In an optimum with full utilization and partial activity, the binding incentive constraint for worker  $i$  at project  $p$  is

$$\alpha_i^p + \sum_{j \neq i, j \in A^p} g_{ji} e_j^{p*} = c_i e_i^{p*} + r_i \sum_{q \neq p, q \in P_i} e_i^{q*}$$

In vector form, we have

$$\boldsymbol{\alpha}^p = \mathbf{M}^p \mathbf{e}^p + (\mathbf{R} \mathbf{E} \mathbf{1}_m)^p$$

where  $(\mathbf{R} \mathbf{E} \mathbf{1}_m)^p$  is the vector of total efforts for workers in  $A^p$  multiplied by  $r_i$ . Solving for  $\mathbf{e}^p$  gives

$$\mathbf{e}^p = (\mathbf{M}^p)^{-1} \boldsymbol{\alpha}^p - (\mathbf{M}^p)^{-1} (\mathbf{R} \mathbf{E} \mathbf{1}_m)^p$$

$\mathbf{M}^p$  is invertible because principal submatrices of positive definite matrices are also positive definite by Sylvester's Criterion. Now I solve for a worker's total effort

$$\mathcal{E}_i = \sum_{p \in P_i} e_i^p = \sum_{p \in P_i} \left( \sum_{j \in A^p} (\mathbf{M}^p)^{-1}_{ij} \alpha_j^p - (\mathbf{M}^p)^{-1}_{ij} r_j \mathcal{E}_j \right)$$

Rearranging, I obtain:

$$\mathcal{E}_i + \sum_{j=1}^n \left( r_j \sum_{p \in P_i \cap P_j} (\mathbf{M}^p)^{-1}_{ij} \right) \mathcal{E}_j = \sum_{p \in P_i} \sum_{j \in A^p} (\mathbf{M}^p)^{-1}_{ij} \alpha_j^p$$

Define  $\hat{\alpha}_i := \sum_{p \in P_i} \sum_{j \in A^p} (\mathbf{M}^p)^{-1}_{ij} \alpha_j^p$  and  $n \times n$  matrix  $\mathcal{P}$  with the following entries:

$$\mathcal{P}_{ij} = \delta_{ij} + r_j \sum_{p \in P_i \cap P_j} (\mathbf{M}^p)^{-1}_{ij}$$

Given  $\mathcal{P}$ , we have the following linear system in workers' total efforts

$$\mathcal{P}(\mathbf{E}\mathbf{1}_m) = \hat{\mathbf{a}}$$

Because  $V$  is strictly concave in total efforts, the optimal total effort is unique and  $\mathcal{P}$  is invertible.

$$\mathbf{E}\mathbf{1}_m = \mathcal{P}^{-1}\hat{\mathbf{a}} \implies \frac{\partial(\mathbf{E}\mathbf{1}_m)}{\partial\gamma} = \mathcal{P}^{-1}\boldsymbol{\sigma}^{(\gamma)}$$

where  $\boldsymbol{\sigma}^{(\gamma)}$  is defined for each model parameter  $\gamma$  as

$$\sigma_i^{(\alpha_j^q)} = \begin{cases} (\mathbf{M}^q)_{ij}^{-1} & \text{if } i \in A^q \\ 0 & \text{o.w.} \end{cases} ; \sigma_i^{(c_j)} = - \sum_{p \in A_i \cap A_j} (\mathbf{M}^p)_{ij}^{-1} e_j^{p*}$$

$$\sigma_i^{(r_j)} = \sum_{p \in A_i \cap A_j} (\mathbf{M}^p)_{ij}^{-1} (e_j^{p*} - \mathcal{E}_j^*) ; \sigma_i^{(g_{i\ell})} = \sum_{p \in A_i \cap A_j \cap A_\ell} (\mathbf{M}^p)_{ij}^{-1} e_\ell^{p*} + (\mathbf{M}^p)_{i\ell}^{-1} e_j^{p*}$$

Now we can finally compute:

$$\frac{\partial \mathbf{e}^p}{\partial \zeta} = \frac{\partial}{\partial \zeta} \left( (\mathbf{M}^p)^{-1} \boldsymbol{\alpha}^p \right) - \frac{\partial}{\partial \zeta} \left( (\mathbf{M}^p)^{-1} (\mathbf{R}\mathbf{E}\mathbf{1}_m)^p \right)$$

Each of the results follows from direct computation of the above equation and substituting in  $\frac{\partial(\mathbf{E}\mathbf{1}_m)}{\partial\zeta} = \mathcal{P}^{-1}\boldsymbol{\sigma}^{(\zeta)}$ .  $\square$

## A.11 Proof of Theorem 7

With underutilization, the necessary and sufficient KKT conditions that  $(\mathbf{e}, \boldsymbol{\lambda})$  satisfy are:

$$\nabla_{\mathbf{e}} V(\mathbf{e}) - \boldsymbol{\lambda} \mathbf{H} = \mathbf{0}$$

$$\forall (i, p) \in \mathcal{F}, \alpha_i^p + \sum_{j \in A^p} g_{ji} e_j^p - c_i e_i^p - r_i \sum_{q \neq p, q \in P_i} e_i^q = 0$$

The corresponding Jacobian is

$$\mathbf{J} = \begin{bmatrix} \nabla_{\mathbf{e}}^2 V(\mathbf{e}) & \mathbf{H}_{\mathcal{F}} \\ \mathbf{H}_{\mathcal{F}}^{\top} & \mathbf{0} \end{bmatrix}$$

Let's show that  $\mathbf{J}$  is invertible. Suppose  $(\mathbf{x}, \mathbf{y}) \in \ker(\mathbf{J})$ , and so

$$\begin{aligned} \nabla_{\mathbf{e}}^2 V(\mathbf{e}^*) \mathbf{x} + (\mathbf{H})_{\mathcal{F}} \mathbf{y} &= \mathbf{0} \\ (\mathbf{H})_{\mathcal{F}}^{\top} \mathbf{x} &= \mathbf{0} \end{aligned}$$

Premultiplying the first equation by  $\mathbf{x}^{\top}$  gives

$$\mathbf{x}^{\top} \nabla_{\mathbf{e}}^2 V(\mathbf{e}^*) \mathbf{x} + \mathbf{x}^{\top} (\mathbf{H})_{\mathcal{F}} \mathbf{y} = 0 \implies \mathbf{x}^{\top} \nabla_{\mathbf{e}}^2 V(\mathbf{e}^*) \mathbf{x} = 0$$

By assumption that  $\nabla_{\mathbf{e}}^2 V(\mathbf{e}^*)$  is strictly negative definite on  $\ker(\mathbf{H}_{\mathcal{F}}^{\top})$ , it must be that  $\mathbf{x} = \mathbf{0}$ . Substituting back into the first equation, we get

$$\mathbf{H}_{\mathcal{F}} \mathbf{y} = \mathbf{0}$$

Since  $\mathbf{H}_{\mathcal{F}}$  is invertible,  $\mathbf{H}_{\mathcal{F}}$  has full row rank and it must be that  $\mathbf{y} = \mathbf{0}$ . This concludes the argument that  $\mathbf{J}$  is invertible. By the Implicit Function Theorem, we obtain the general formula for how  $(\mathbf{e}^*, \lambda^*)$  react to a change in a parameter  $\theta$ :

$$\begin{pmatrix} \frac{\partial \mathbf{e}^*}{\partial \theta} \\ \frac{\partial \lambda^*}{\partial \theta} \end{pmatrix} = -\mathbf{J}^{-1} \begin{pmatrix} \nabla_{\mathbf{e}, \theta}^2 V(\mathbf{e}^*) - \nabla_{\theta}(\lambda(\mathbf{H})) \\ \nabla_{\theta}(\boldsymbol{\alpha} - \mathbf{H}_{\mathcal{F}}) \end{pmatrix}$$

Next, I compute the right-hand vectors associated with each model parameter.

$$\begin{aligned} \mathbf{a}_i^p &= \begin{pmatrix} \mathbf{0} \\ \boldsymbol{\eta}_i^p \cdot \lambda_i^{p*} \end{pmatrix} ; \quad \mathbf{g}_{ji} = \begin{pmatrix} \boldsymbol{\epsilon}_i^p \cdot \lambda_j^{p*} + \boldsymbol{\epsilon}_j^p \cdot \lambda_i^{p*} \\ -\boldsymbol{\eta}_i^p \cdot (e_j^p) - \boldsymbol{\eta}_j^p \cdot (e_i^p) \end{pmatrix} \\ \mathbf{c}_i &= \begin{pmatrix} -\sum_{p \in P_i} \boldsymbol{\epsilon}_i^p \cdot \lambda_i^{p*} \\ \sum_{p \in P_i} \boldsymbol{\eta}_i^p \cdot e_i^{p*} \end{pmatrix} ; \quad \mathbf{r}_i = \begin{pmatrix} -\sum_{p \in P_i} \boldsymbol{\epsilon}_i^p \cdot \sum_{q \neq p} \lambda_i^{q*} \\ \sum_{p \in P_i} \boldsymbol{\eta}_i^p \cdot \sum_{q \neq p} e_i^{q*} \end{pmatrix} \end{aligned}$$

To get the final formulas in the theorem, multiply the first row of  $\mathbf{J}^{-1}$  by the corresponding right-hand vector for each model parameter.  $\square$

## A.12 Proof of Proposition 5

Worker  $i$ 's objective,  $U_i(\mathbf{e}_i, \mathbf{e}_{-i}) + w_i(\mathbf{e}_i)$ , is strictly concave by [Assumption 1](#), so  $\mathbf{e}_i$  is a best response to all other workers efforts if and only if  $b_i^p + \alpha_i^p + \sum_{j \neq i} g_{ij} e_j^p - c_i e_i^p - r_i \sum_{q \neq p} e_i^q \leq 0$  for all projects  $p$ , and holds with equality if  $e_i^p > 0$ . It is the unique equilibrium because negative definiteness of  $\mathbf{H}$  remains with the addition of wages.  $\square$

## B Generic Conditions in the Optimum

Statement (1) is the usual strict complementary slackness condition on  $\lambda$ , which is generic for convex optimization problems that satisfy strict feasibility.

To show that statement (2),  $e_i^p = 0$  implies  $\lambda_i^p = 0$ , is a generic condition, I argue that the set of model parameters that satisfy [Assumption 2](#) with  $e_i^p = 0$  and a binding incentive constraint for some worker-project pair  $(i, p)$  has measure zero. I label this set as  $\mathcal{M}$ . Crucially, this *does not* imply the converse i.e. that  $\lambda_i^p = 0$  implies  $e_i^p = 0$ . In the environment with homogeneous workers, underutilizing workers, i.e.,  $\lambda_i^p = 0$  with  $e_i^p > 0$ , is a generic feature of the optimal project assignment.

I write  $(\alpha, \mathbf{c}, \mathbf{g}, \mathbf{r})$  to be an instance of the model parameters. First, note that the planner's objective is strictly concave in total efforts, the feasible set is convex, Slater's condition holds, and the optimal effort totals  $\mathcal{E}^p(\alpha, \mathbf{c}, \mathbf{g}, \mathbf{r})$  and  $\mathcal{E}_i(\alpha, \mathbf{c}, \mathbf{g}, \mathbf{r})$  are unique. On an open, dense subset  $\mathcal{U}$  of the space of valid model parameters, the Implicit Function Theorem implies  $\mathcal{E}^p(\alpha, \mathbf{c}, \mathbf{g}, \mathbf{r})$  and  $\mathcal{E}_i(\alpha, \mathbf{c}, \mathbf{g}, \mathbf{r})$  are real-analytic functions. Assume by contradiction that  $\mathcal{M}$  has a non-empty interior. This implies the subset  $\mathcal{M}_i^p(\alpha, \mathbf{c}, \mathbf{g}, \mathbf{r}) := \alpha_i^p + g \mathcal{E}^p(\alpha, \mathbf{c}, \mathbf{g}, \mathbf{r}) - r_i \mathcal{E}_i(\alpha, \mathbf{c}, \mathbf{g}, \mathbf{r})$  vanishes on an open set within  $\mathcal{U}$  for some  $(i, p)$ . By the Identity Theorem for real-analytic functions,  $\mathcal{M}_i^p(\alpha, \mathbf{c}, \mathbf{g}, \mathbf{r}) \equiv 0$  everywhere on  $\mathcal{U}$ .

By continuity, there exists some  $\epsilon > 0$  sufficiently small such that  $r_j = \epsilon$  for



all workers implies every worker is fully active and fully utilized at every project. With vanishing cross-project costs, each worker treats each project independently. Since  $V$  is strictly increasing in each argument, the principal assigns each worker to each project and pushes them to exert the highest feasible effort, made possible by fully utilizing them. Thus,  $\mathcal{M}_i^p(\alpha, \mathbf{c}, \mathbf{g}, \mathbf{r}) > 0$  for a set of model parameters  $(\alpha, \mathbf{c}, \mathbf{g}, \mathbf{r})$  with positive measure. Thus  $\mathcal{M}_i^p$  cannot contain an open set and must be measure zero. Thereby,  $\mathcal{M} = \cup_{i,p} \mathcal{M}_i^p$  also has measure zero. We conclude that  $e_i^p = 0$  implies  $\lambda_i^p = 0$  is a generic property of the solution and can be assumed without loss of generality.

## C The General MINLP

The principal's general problem is written as a mixed-integer nonlinear program (MINLP).

$$\begin{aligned}
& \max_{\mathbf{e}, \mathbf{z}} V(\mathcal{E}^1, \dots, \mathcal{E}^m) \\
& \text{s.t. } \mathbf{e} \geq \mathbf{0}, \\
& \mathbf{z} \in \{0, 1\}^{mn}, \\
& \forall(i, p), e_i^p \leq Mz_i^p \\
& \forall(i, p), \alpha_i^p + \sum_{j \neq i} g_{ij} e_j^p - c_i e_i^p - r_i \sum_{q \neq p} e_i^q \geq -M(1 - z_i^p) \\
& \forall p, \sum_{i=1}^n e_i^p = \mathcal{E}^p
\end{aligned} \tag{P3}$$

where  $M$  is a sufficiently large positive constant.<sup>24</sup>  $z_i^p$  is a binary variable that acts as a switch to ensure that if  $e_i^p > 0$ , then  $\frac{\partial B_i^p}{\partial e_i^p}(\mathbf{e}) - \frac{\partial C_i}{\partial e_i^p}(\mathbf{e}_i) \geq 0$ . If that is the case, then  $z_i^p$  must be 1. Otherwise, if  $e_i^p = 0$ , then no incentive constraint is needed and  $z_i^p$  is set to 0. Switching the incentive constraints turns the principal's problem into a mixed-integer nonlinear programming problem, rendering it *NP*-hard.

<sup>24</sup>For example, one can set  $M = \|(\mathbf{C} \otimes \mathbf{I}_m - \mathbf{G} \otimes \mathbf{I}_m)^{-1} \alpha\|_\infty$ .  $(\mathbf{C} \otimes \mathbf{I}_m - \mathbf{G} \otimes \mathbf{I}_m)^{-1} \alpha$  gives the vector of maximum efforts workers could play at each project, absent any substitutability across projects. Instead of using a Big- $M$  constant, we can also write the principal's problem with disjunctive constraints: if  $z_i^p = 1$ , then  $\alpha_i^p + \sum_{j \neq i} g_{ij} e_j^p - c_i e_i^p - r_i \sum_{q \neq p} e_i^q \geq 0$ ; else  $e_i^p = 0$ .

To see an explicit example of parameters where the feasible set is nonconvex, take a simple example with one worker  $i$  and two projects,  $p$  and  $q$ . Let the productivity parameters be  $\alpha_i^p = 4$ ,  $\alpha_i^q = 1$ ,  $c_i = 0.5$ ,  $r_i = 0.3$ . He prefers working on project  $p$  four times as much as on project  $q$ . The feasible set contains all pair of nonnegative efforts  $(e_i^p, e_i^q)$  that satisfy:

$$\begin{aligned} 4 &\geq 0.5e_i^p + 0.3e_i^q && \text{if } e_i^p > 0 \\ 1 &\geq 0.5e_i^q + 0.3e_i^p && \text{if } e_i^q > 0 \end{aligned}$$

Figure 8 illustrates the boundary on the set of implementable efforts. Effort vectors of the form  $(e_i^p, 0)$  for  $e_i^p \in [3\frac{1}{3}, 4]$  are feasible, as  $0.5e_i^p \leq 2$ .  $e_i^q$  is zero at those efforts, and so his incentive constraint for project  $q$  has no bite.

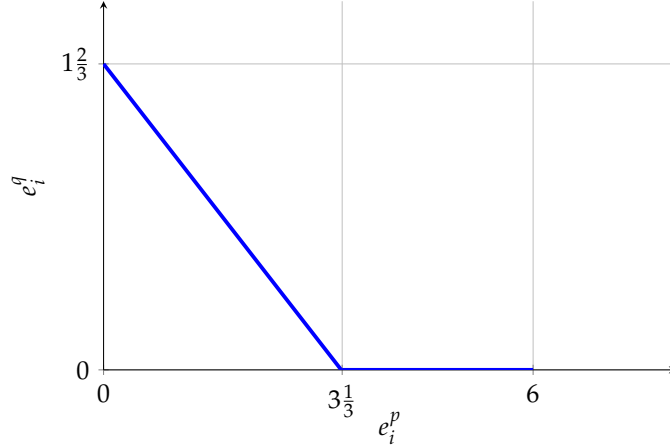


Figure 8: Nonconvex Boundary of Implementable Efforts

The inherent difficulty in analyzing the MINLP comes from the combinatorial nature of the problem. Traditional approaches such as using a greedy algorithm fail because it is neither monotone nor submodular. Increasing the effort cap for a worker at a particular project may cause him to reduce efforts at other projects, leading to an overall decrease in the principal's valuation. Raising effort caps could also cause other workers to increase their efforts on that particular project due to complementarity, increasing the principal's valuation in a convex fashion.

Algorithmic methods, such as outer approximations, serve as efficient ways to compute the solution to the MINLP. Unfortunately, algorithmic approaches provide little in terms of structural and economic insights. Alternatively, absent the

sufficient condition on the model parameters in [Proposition 2](#), we can interpret the comparative statics exercises as local analysis treating the binary  $\mathbf{z}$  variables in the MINLP as fixed.